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CUSTOMER CHURN ANALYSIS

for a Banking Industry client

A COMPREHENSIVE REPORT

Making money empowering

For most digitally native Indians, managing money comes with hidden fees, jargon and stress.



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EXECUTIVE SUMMARY

This report presents a SQL-based customer churn analysis conducted on a banking dataset containing 10,000 customer records. The objective of the analysis was to identify the key factors influencing customer attrition and determine the most effective metrics for understanding churn behavior.

Using PostgreSQL, exploratory data analysis, summary statistics, and multi-variable segmentation techniques were performed across demographic, financial, and behavioral attributes including age, geography, gender, balance, credit score, number of products, tenure, active membership status, estimated salary, and credit card ownership.

The analysis revealed an overall customer churn rate of **20.37%**, with significant variation across different customer segments.

The strongest churn indicators identified in the dataset were:

- **Age**
- **Geography**
- **Number of Products**
- **Active Membership Status**

The analysis showed that:

- Customers aged **40–69** demonstrated significantly **higher churn** tendencies
- **Germany** consistently recorded the **highest churn rates** across multiple segments
- Customers with only **1 product** or excessive product ownership (**3–4 products**) exhibited **elevated churn**
- **Inactive customers** were substantially **more likely to exit** compared to active members

Moderate influence on churn behavior was observed for:

- Gender
- Balance
- Credit Score

Meanwhile:

- Estimated Salary
- Tenure
- Credit Card Ownership

showed relatively weak relationships with customer attrition.

Combined attribute analysis further highlighted that churn risk is concentrated within specific high-risk customer groups, particularly middle-aged customers in Germany and inactive customers with low engagement levels.

Based on these findings, several business recommendations were proposed, including:

- improving engagement for inactive users,
- strengthening retention efforts in Germany,
- encouraging balanced product adoption,
- and implementing segmentation-based customer retention strategies.

Overall, the report demonstrates how SQL-driven business analytics can support data-informed decision-making and help financial institutions identify high-risk customer segments, improve customer retention, and enhance long-term customer relationship management.



ABOUT JUPITER MONEY

What comes to mind when you think of traditional banking? Long queues, extensive paperwork, and time-consuming processes. Jupiter Money aims to eliminate these inefficiencies by offering a seamless and digital-first banking experience.

Jupiter is an all-in-one money management application designed to simplify personal finance. Its core objective is to deliver a financial experience that keeps pace with modern customers by providing intuitive tools, smart insights based on spending behavior, and innovative features that help users better understand and manage their finances.

MISSION, VISION, AND GOALS

Jupiter's mission is to improve customers' financial well-being by providing complete control over their money—enabling them to track, save, and invest with confidence.

Built for a digital-first India, Jupiter focuses on simplicity, transparency, and user empowerment. The platform is designed to align with fast-paced lifestyles, ensuring that financial services are accessible, efficient, and easy to use.

Trust and Security Framework

To ensure reliability and customer trust:

- Savings accounts are hosted in partnership with Federal Bank
- Customer data is protected through end-to-end encryption and complies with ISO and PCI security standards
- Deposits are insured up to ₹5,00,000 under Deposit Insurance and Credit Guarantee Corporation (DICGC)



THE CUSTOMER CHURN PROBLEM

In today's competitive banking landscape, customers have access to a wide range of financial service providers, making it easier than ever to switch between banks. This increased competition presents a significant challenge for institutions in retaining customers.

From a business perspective, customer churn directly impacts profitability, customer lifetime value, and long-term growth. Therefore, it is critical to identify the factors that influence a customer's decision to discontinue services.

This report leverages SQL-based data analysis to examine customer behavior patterns and identify key drivers of churn. The insights derived aim to support data-driven decision-making and help the organization design effective customer retention strategies.

PROBLEM STATEMENT

The objective of this report is to analyze customer churn pattern in a banking client dataset and identify key metrics influencing customer exit using SQL-based analysis.

DATASET OVERVIEW

The dataset used in this analysis consists of **10,000 customer records**, where each row represents an individual bank customer identified uniquely using *customerid*, which has been defined as the primary key in the SQL database table **bank_churn**.

The dataset contains a mix of demographic, financial, and behavioral attributes that help in understanding customer profiles and identifying patterns related to churn.

Dataset Structure

The key features included in the dataset are as follows:

- *RowNumber*: Sequential identifier for each record
- *CustomerId*: Unique identifier for each customer (Primary Key)
- *Surname*: Customer's last name
- *CreditScore*: Creditworthiness score assigned to the customer
- *Geography*: Country of residence
- *Gender*: Customer gender
- *Age*: Age of the customer
- *Tenure*: Number of years the customer has been with the bank
- *Balance*: Account balance maintained by the customer
- *NumOfProducts*: Number of banking products held by the customer
- *HasCrCard*: Indicates whether or not the customer holds a credit card (Boolean)
- *IsActiveMember*: Indicates customer activity status (Boolean)
- *EstimatedSalary*: Estimated annual income of the customer
- *Exited*: Target variable indicating customer churn status (Boolean)

Target Variable: Customer Churn (Exited)

The primary variable of interest in this analysis is *Exited*, which indicates whether or not a customer has churned (left the bank) or stayed as a customer with the bank.

To understand the distribution of churn within the dataset, SQL aggregation was performed:

- **Churned customers (Exited = TRUE)**
 - **Retained customers (Exited = FALSE)**
- ```
COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count
```

Additionally, **churn percentage** was calculated using SQL as:

```
ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS "exited
percentage ((exit/exit+stay)*100)"
```

This helps in understanding the proportion of customers who have discontinued services relative to the total customer base.

### Key Insight from Dataset Structure

The dataset is well-suited for churn analysis as it includes:

- Demographic variables (age, gender, geography)
- Financial indicators (balance, credit score, salary)
- Behavioral indicators (tenure, product usage, activity status)

These features allow for multi-dimensional analysis of churn behavior using SQL-based segmentation and aggregation techniques.

| RowNumber | CustomerId | Surname     | CreditScore | Geography | Gender | Age | Tenure | Balance   | NumOfProducts | HasCrCard | IsActiveMember | EstimatedSalary | Exited |
|-----------|------------|-------------|-------------|-----------|--------|-----|--------|-----------|---------------|-----------|----------------|-----------------|--------|
| 9990      | 15605622   | McMillan    | 841         | Spain     | Male   | 28  | 4      | 0         | 2             | 1         | 1              | 179436.6        | 0      |
| 9991      | 15798964   | Nkemakonam  | 714         | Germany   | Male   | 33  | 3      | 35016.6   | 1             | 1         | 0              | 53667.08        | 0      |
| 9992      | 15769959   | Ajuluchukwu | 597         | France    | Female | 53  | 4      | 88381.21  | 1             | 1         | 0              | 69384.71        | 1      |
| 9993      | 15657105   | Chukwualuka | 726         | Spain     | Male   | 36  | 2      | 0         | 1             | 1         | 0              | 195192.4        | 0      |
| 9994      | 15569266   | Rahman      | 644         | France    | Male   | 28  | 7      | 155060.41 | 1             | 1         | 0              | 29179.52        | 0      |
| 9995      | 15719294   | Wood        | 800         | France    | Female | 29  | 2      | 0         | 2             | 0         | 0              | 167773.55       | 0      |
| 9996      | 15606229   | Obijaku     | 771         | France    | Male   | 39  | 5      | 0         | 2             | 1         | 0              | 96270.64        | 0      |
| 9997      | 15569892   | Johnstone   | 516         | France    | Male   | 35  | 10     | 57369.61  | 1             | 1         | 1              | 101699.77       | 0      |
| 9998      | 15584532   | Liu         | 709         | France    | Female | 36  | 7      | 0         | 1             | 0         | 1              | 42085.58        | 1      |
| 9999      | 15682355   | Sabbatini   | 772         | Germany   | Male   | 42  | 3      | 75075.31  | 2             | 1         | 0              | 92888.52        | 1      |
| 10000     | 15628319   | Walker      | 792         | France    | Female | 28  | 4      | 130142.79 | 1             | 1         | 0              | 38190.78        | 0      |

## METHODOLOGY USED

This section outlines the analytical approach used to explore the dataset, perform SQL-based analysis, and identify key drivers of customer churn. The methodology is divided into three stages: data exploration, summary statistics, and metric selection.

### **a) Data Exploration (DDL & DML)**

The initial phase involved exploring the dataset using SQL-based DDL and DML operations to understand customer segmentation and churn behavior.

#### Level 1: Basic Segmentation

Customers were first analyzed based on overall churn distribution and individual attributes:

- Gender
- Geography
- Credit Score
- Balance
- Age
- Estimated Salary
- Active Membership Status (IsActiveMember)
- Number of Products
- Tenure
- Credit Card Ownership (HasCrCard)

Key comparisons were made using:

- Exited Count vs Stayed Count
- Exited Percentage

### Level 2: Combined Attribute Analysis

To understand interaction effects between variables, deeper segmentation was performed using multi-variable grouping techniques. This enabled analysis of churn behavior across combinations of demographic, financial, and behavioral attributes rather than isolated variables alone.

The following combinations were explored as part of the analysis:

- Gender + Credit Score
- Gender + Geography
- Gender + Age
- Balance + Geography
- Balance + Geography + Salary
- Age + Geography
- Age + Geography + Salary
- Geography + Credit Score
- Geography + Salary
- Geography + Age + Balance + Salary
- Geography + Gender + Age + Balance
- Geography + Gender + Age + Balance + Salary

This stage of analysis helped identify deeper churn patterns, interaction effects between variables, and high-risk customer segments that may not be visible through single-variable analysis alone.

The most relevant, actionable, and business-significant findings from the combined attribute analysis are discussed in further detail within the report. Additional exploratory outputs and supporting tables have been included in the appendix section at the very end of the report.

### Level 3: Country-wise Analysis

A deeper geographic analysis was conducted across the three countries present in the dataset:

- France
- Spain
- Germany

Within each country, customer churn was further explored across multiple variables, including:

- Age
- Balance
- Credit Score
- Gender
- Number of Products
- Estimated Salary

This country-wise segmentation helped identify regional variations in customer behavior and differences in churn trends across demographic and financial categories.

However, since the Combined Attribute Analysis section already covers the most relevant and actionable geography-based churn patterns in detail, the additional country-wise exploratory analyses and supporting segmentation outputs have been included in the appendix to maintain report clarity and avoid repetitive interpretation.

### **b) Summary Statistics**

To understand the distribution and variability of numerical features, summary statistics were calculated using SQL aggregate functions.

The following metrics were computed for key variables:

- Mean
- Median
- Mode
- Minimum
- Maximum
- Standard Deviation

#### **Variables analyzed:**

- Age
- Balance
- Credit Score
- Estimated Salary
- Number of Products
- Tenure

#### **Purpose of Analysis**

This step helped assess:

- Data distribution patterns
- Presence of skewness or outliers
- Variation across financial and demographic attributes

### **c) Metrics Selection**

Based on exploratory and statistical analysis, key metrics influencing customer churn were identified.

#### **Strong Influencing Factors**

The following variables showed significant correlation with churn behavior:

1. **Number of Products**
2. **Active Membership Status**
3. **Age**
4. **Geography**

These factors were observed to create distinct differences between churned and retained customers.

#### **Moderate Influencing Factors**

The following variables showed a moderate relationship with churn behavior:

- **Gender**
- **Balance**
- **Credit Score**

These variables demonstrated noticeable differences in churn percentages across customer groups; however, the variation was not strong or consistent enough to classify them as dominant standalone churn indicators. Their impact becomes more meaningful when analyzed in combination with other demographic, financial, and behavioral factors.

### **Weak or Non-Significant Factors**

Some variables showed limited or inconsistent impact on churn:

- **Estimated Salary**

Churn percentages remained relatively stable across all salary ranges, generally fluctuating between 18% and 23%. Since both exited and retained customers were distributed consistently across income groups, estimated salary does not appear to strongly influence customer churn behavior.

- **Tenure**

Churn distribution remained relatively uniform across all tenure groups, indicating that customer duration with the bank does not significantly influence exit behavior.

- **Credit Card Ownership**

Customers with and without credit cards exhibited nearly identical churn rates:

- Customers with credit cards: ~20.18% churn
- Customers without credit cards: ~20.81% churn

Since the variation is minimal, credit card ownership is not a strong differentiator for churn prediction.

### **Key Observation**

The analysis indicates that churn is driven by a combination of demographic and financial attributes rather than any single dominant variable. This highlights the importance of multi-dimensional segmentation for accurate churn prediction.

## SQL ANALYSIS & KEY FINDINGS

### A) Data Exploration LEVEL 1: Basic Segmentation

#### 1. Overall Customer Churn Rate

The initial analysis focused on understanding the overall churn distribution in the dataset. Out of 10,000 customers, a total of 2,037 customers have exited the banking service, representing a churn rate of 20.37%. This indicates that approximately 1 in every 5 customers has discontinued using the bank's services.

SQL Query Used:

```
select count (*) filter (where exited = true) as total_customers_churned,
 count (*) as total_customers,
 ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2)
 AS "churn percentage"
from bank_churn order by "churn percentage" desc;
```

|   | total_customers_churned<br>bigint | total_customers<br>bigint | churn percentage<br>numeric |
|---|-----------------------------------|---------------------------|-----------------------------|
| 1 | 2037                              | 10000                     | 20.37                       |

#### Insight

The churn rate of **20.37%** indicates a significant customer attrition level for the bank. This suggests that a considerable portion of the customer base is not being retained, highlighting the need for deeper analysis into behavioral, financial, and demographic factors influencing customer exit.

#### 2. Customer Churn by Geography 🌐

To understand regional variations in customer attrition, churn was analyzed across the three countries present in the dataset: France, Germany, and Spain.

SQL Query Used:

```
select geography,
 COUNT(*) FILTER (WHERE exited = TRUE) AS exited_customers,
 COUNT(*) FILTER (WHERE exited = FALSE) AS retained_customers,
 COUNT (*) AS total_customers,
 ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage
from bank_churn group by geography order by churn_percentage desc;
```

|   | geography<br>character varying (20) | exited_customers<br>bigint | retained_customers<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|---|-------------------------------------|----------------------------|------------------------------|---------------------------|-----------------------------|
| 1 | Germany                             | 814                        | 1695                         | 2509                      | 32.44                       |
| 2 | Spain                               | 413                        | 2064                         | 2477                      | 16.67                       |
| 3 | France                              | 810                        | 4204                         | 5014                      | 16.15                       |

#### Insight

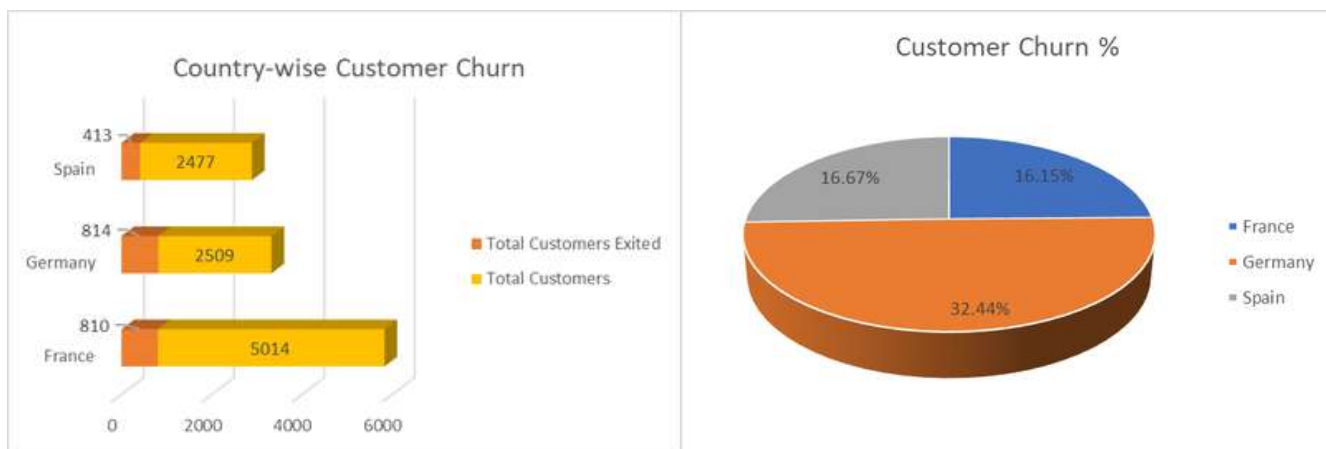
The analysis reveals a significant regional disparity in churn behavior:

**Germany exhibits the highest churn rate at 32.44%**, which is almost double compared to France and Spain.

In contrast, **France (16.15%) and Spain (16.67%) show relatively stable and similar churn levels.**

Despite having a smaller customer base than France, Germany contributes disproportionately to overall churn, making it the most critical region from a retention perspective.

A bar chart was used to compare total vs exited customers across countries



A pie chart was used to represent churn percentage distribution across geographies

### Business Interpretation

This indicates that geography plays a strong role in customer retention patterns. Customers in Germany are significantly more likely to exit the bank compared to other regions, suggesting potential issues such as:

- Differences in customer expectations
- Market competition intensity
- Product-market fit in that region

From a business standpoint, Germany should be prioritized for targeted retention strategies and deeper behavioral analysis.

### 3. Customer Churn by Age Group

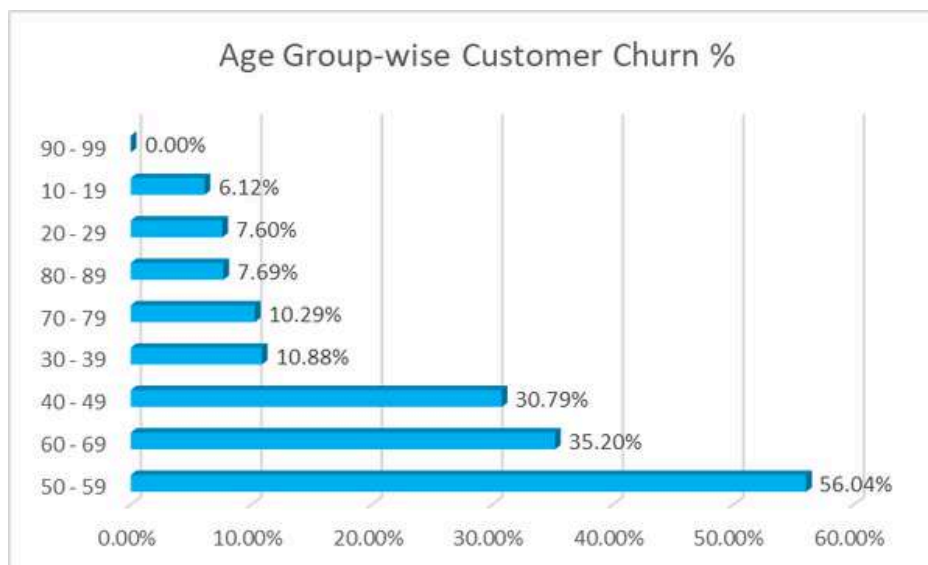
To understand the relationship between customer age and churn behavior, customers were segmented into age bands of 10-year intervals. This allows identification of behavioral patterns across different life stages.

SQL Query Used:

```
SELECT
 CONCAT(FLOOR(age/10) * 10, ' - ', FLOOR(age/10) * 10 + 9) AS age_range,
 COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
 COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count,
 COUNT(*) AS total_customers,
 ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2)
 AS churn_percentage
FROM bank_churn WHERE age <= 100 GROUP BY FLOOR(age/10)
ORDER BY churn_percentage DESC;
```

|   | age_range<br>text | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|---|-------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1 | 50 - 59           | 487                    | 382                      | 869                       | 56.04                       |
| 2 | 60 - 69           | 132                    | 243                      | 375                       | 35.20                       |
| 3 | 40 - 49           | 806                    | 1812                     | 2618                      | 30.79                       |
| 4 | 30 - 39           | 473                    | 3873                     | 4346                      | 10.88                       |
| 5 | 70 - 79           | 14                     | 122                      | 136                       | 10.29                       |
| 6 | 80 - 89           | 1                      | 12                       | 13                        | 7.69                        |
| 7 | 20 - 29           | 121                    | 1471                     | 1592                      | 7.60                        |
| 8 | 10 - 19           | 3                      | 46                       | 49                        | 6.12                        |
| 9 | 90 - 99           | 0                      | 2                        | 2                         | 0.00                        |

A bar chart was used to compare churn percentage across age groups



This clearly highlights the sharp increase in churn among middle-aged customers

### Insight

Age shows a strong relationship with customer churn behavior.

The analysis reveals a clear concentration of churn in the **40–69 age segment**, with the highest churn observed in the **50–59 group (56.04%)**, followed by the **60–69 group (35.20%)**, and the **40–49 group (30.79%)**.

In contrast, younger customers (10–39 years) exhibit significantly lower churn rates, with the 30–39 age group showing a relatively stable churn rate of 10.88%, the 10–19 group at 6.12%, and the 20–29 group at 7.60%. Similarly, super-senior citizens (70–89) also show significantly lower churn rates, with the 70–79 group at 10.29%, and the 80–89 group at 7.69%.

### Business Interpretation

This indicates that **middle-aged customers are the most vulnerable segment for churn**.

Possible explanations include:

- Higher financial expectations in mid-life stages
- Increased comparison of banking services
- Greater sensitivity to product value and returns

Meanwhile, younger customers tend to show lower churn, possibly due to:

- Lower financial complexity
- Fewer alternative banking commitments
- Early-stage relationship with the bank

### Key Takeaway

**Age is a strong predictive factor for churn**, with a clear non-linear pattern:

- Churn peaks in middle age (40–69)
- Drops significantly in younger and older segments

This makes age one of the **primary segmentation variables for churn prediction modeling and targeted retention strategies**.

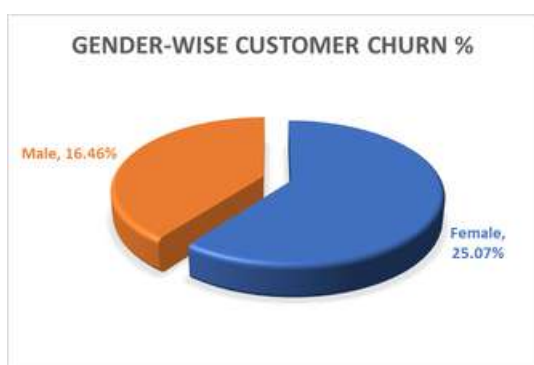
#### 4. Customer Churn by Gender 🧑

To analyze the relationship between gender and customer churn behavior, customers were segmented into male and female categories and compared based on exited and retained counts.

SQL Query Used:

```
SELECT gender, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, COUNT(*) AS total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn GROUP BY gender ORDER BY churn_percentage DESC;
```

|   | gender<br>character varying (10) | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|---|----------------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1 | Female                           | 1139                   | 3404                     | 4543                      | 25.07                       |
| 2 | Male                             | 898                    | 4559                     | 5457                      | 16.46                       |



A pie chart was used to compare churn percentages across gender categories



Additional comparison was made between exited and total customer counts for each group

#### Insight

The analysis indicates a noticeable difference in churn behavior between male and female customers.

Female customers exhibit a significantly higher churn rate of 25.07%, compared to 16.46% among male customers. Although the total number of male customers in the dataset is higher, female customers demonstrate a greater likelihood of discontinuing banking services.

#### Business Interpretation

This suggests that gender may influence customer retention behavior and customer experience expectations.

The higher churn observed among female customers may indicate:

- Lower satisfaction with existing banking services
- Differences in product engagement patterns
- Variations in financial behavior and service expectations

However, gender alone may not fully explain churn behavior and should be analyzed alongside other variables such as age, balance, geography, and product usage for more accurate insights.

#### Key Takeaway

Gender appears to be a **moderately strong factor** influencing churn in this dataset, with female customers showing considerably higher attrition rates than male customers.

This highlights the importance of designing more personalized customer engagement and retention strategies across demographic segments.

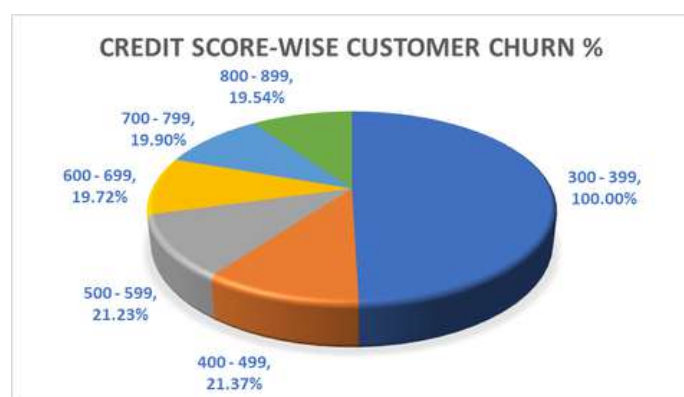
## 5. Customer Churn by Credit Score

To evaluate the impact of financial credibility on customer churn, customers were segmented into credit score ranges of 100-point intervals and analyzed based on churn distribution.

SQL Query Used:

```
SELECT CONCAT(FLOOR(creditscore / 100) * 100, ' - ', FLOOR(creditscore / 100) * 100 + 99) AS creditscore_range,
COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count,
COUNT(*) AS total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2)
AS churn_percentage FROM bank_churn WHERE creditscore <= 1000 GROUP BY FLOOR(creditscore / 100)
ORDER BY churn_percentage DESC;
```

|   | creditscore_range<br>text | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|---|---------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1 | 300 - 399                 | 19                     | 0                        | 19                        | 100.00                      |
| 2 | 400 - 499                 | 131                    | 482                      | 613                       | 21.37                       |
| 3 | 500 - 599                 | 510                    | 1892                     | 2402                      | 21.23                       |
| 4 | 700 - 799                 | 496                    | 1997                     | 2493                      | 19.90                       |
| 5 | 600 - 699                 | 753                    | 3065                     | 3818                      | 19.72                       |
| 6 | 800 - 899                 | 128                    | 527                      | 655                       | 19.54                       |



A pie chart was used to compare churn percentages across credit score ranges



Additional comparison was made between exited and total customer counts for each group

### Insight

The analysis shows that churn percentages remain relatively consistent across most credit score categories, **generally ranging between 19% and 21%.**

The **300–399 credit score range shows a 100% churn rate**; however, this segment contains **only 19 customers**, making it statistically insignificant for broader business conclusions.

Among larger customer groups:

- Customers in the **400–599** range exhibit slightly higher churn rates (~21%)
- Customers with higher credit scores (600–899) show marginally lower churn levels (~19.5%)

Overall, the variation in churn across major credit score ranges is relatively small.

### Business Interpretation

Credit score appears to have a **moderate influence** on customer churn rather than acting as a dominant predictive factor.

While customers with lower credit scores display slightly higher churn tendencies, the differences are not substantial enough to indicate that financial creditworthiness alone drives customer exit decisions.

This suggests that churn behavior is more likely influenced by a combination of customer engagement, banking product usage, geographic factors, and financial behavior patterns.

## Key Takeaway

Credit score demonstrates a **limited-to-moderate relationship** with customer churn in this dataset.

Although lower credit score groups show somewhat elevated churn rates, the overall churn distribution remains relatively stable across major score categories. Therefore, credit score should be considered a **supporting metric** rather than a primary standalone churn indicator.

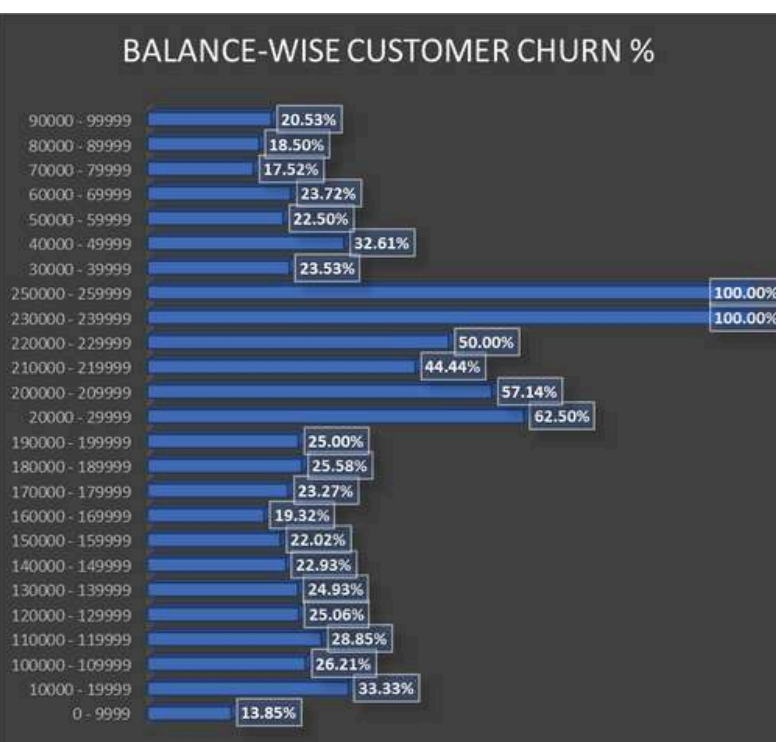
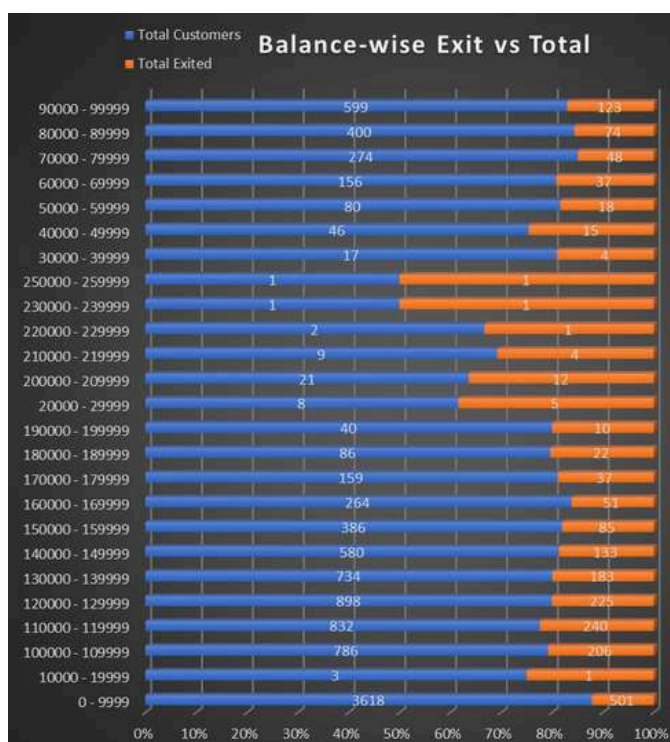
## 6. Customer Churn by Account Balance 💰

To understand the relationship between customer account balance and churn behavior, customers were grouped into balance ranges of ₹10,000 intervals and analyzed based on exited and retained customer counts.

SQL Query Used:

```
SELECT CONCAT(FLOOR(balance / 10000) * 10000, ' - ', FLOOR(balance / 10000) * 10000 + 9999) AS balance_range,
COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count,
COUNT(*) AS total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2)
AS churn_percentage FROM bank_churn WHERE balance <= 300000 GROUP BY FLOOR(balance / 10000)
ORDER BY churn_percentage DESC;
```

|    | balance_range<br>text | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|-----------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | 250000 - 259999       | 1                      | 0                        | 1                         | 100.00                      |
| 2  | 230000 - 239999       | 1                      | 0                        | 1                         | 100.00                      |
| 3  | 20000 - 29999         | 5                      | 3                        | 8                         | 62.50                       |
| 4  | 200000 - 209999       | 12                     | 9                        | 21                        | 57.14                       |
| 5  | 220000 - 229999       | 1                      | 1                        | 2                         | 50.00                       |
| 6  | 210000 - 219999       | 4                      | 5                        | 9                         | 44.44                       |
| 7  | 10000 - 19999         | 1                      | 2                        | 3                         | 33.33                       |
| 8  | 40000 - 49999         | 15                     | 31                       | 46                        | 32.61                       |
| 9  | 110000 - 119999       | 240                    | 592                      | 832                       | 28.85                       |
| 10 | 100000 - 109999       | 206                    | 580                      | 786                       | 26.21                       |
| 11 | 180000 - 189999       | 22                     | 64                       | 86                        | 25.58                       |
| 12 | 120000 - 129999       | 225                    | 673                      | 898                       | 25.06                       |
| 13 | 190000 - 199999       | 10                     | 30                       | 40                        | 25.00                       |
| 14 | 130000 - 139999       | 183                    | 551                      | 734                       | 24.93                       |
| 15 | 60000 - 69999         | 37                     | 119                      | 156                       | 23.72                       |
| 16 | 30000 - 39999         | 4                      | 13                       | 17                        | 23.53                       |
| 17 | 170000 - 179999       | 37                     | 122                      | 159                       | 23.27                       |
| 18 | 140000 - 149999       | 133                    | 447                      | 580                       | 22.93                       |
| 19 | 50000 - 59999         | 18                     | 62                       | 80                        | 22.50                       |
| 20 | 150000 - 159999       | 85                     | 301                      | 386                       | 22.02                       |
| 21 | 90000 - 99999         | 123                    | 476                      | 599                       | 20.53                       |
| 22 | 160000 - 169999       | 51                     | 213                      | 264                       | 19.32                       |
| 23 | 80000 - 89999         | 74                     | 326                      | 400                       | 18.50                       |
| 24 | 70000 - 79999         | 48                     | 226                      | 274                       | 17.52                       |
| 25 | 0 - 9999              | 501                    | 3117                     | 3618                      | 13.85                       |



Additional comparison was made between exited and total customer counts for each group

A bar chart was used to compare churn percentages across balance ranges

### Insight

The analysis shows varying churn behavior across account balance segments. Several extremely high churn percentages (50%–100%) are observed in balance ranges above ₹200,000; however, these segments contain very few customers and therefore are not statistically reliable for broader conclusions.

Among the larger customer groups:

- Customers with balances between ₹90,000 and ₹199,999, and ₹50,000–₹69,999, show moderate to relatively elevated churn rates ranging between 19% and 29%
- The ₹0–₹9,999 segment, despite having the highest customer volume (3,618 customers), shows a comparatively lower churn rate of 13.85%

### Business Interpretation

The findings suggest that account balance has a **moderate relationship** with customer churn. Customers maintaining medium-to-high balances demonstrate somewhat higher churn tendencies compared to customers with very low balances. This may indicate that higher-value customers have:

- greater banking expectations,
- access to competing financial institutions,
- or increased sensitivity to service quality and benefits.

At the same time, the retained customer count remains high across most balance ranges, indicating that balance alone cannot fully explain churn behavior.

### Key Takeaway

Balance should be considered a **supporting churn indicator rather than a dominant standalone metric**.

While certain balance segments exhibit elevated churn percentages, the relationship is not sufficiently consistent across all customer groups. Therefore, balance is more effective when analyzed in combination with other variables such as geography, age, product usage, and activity status.

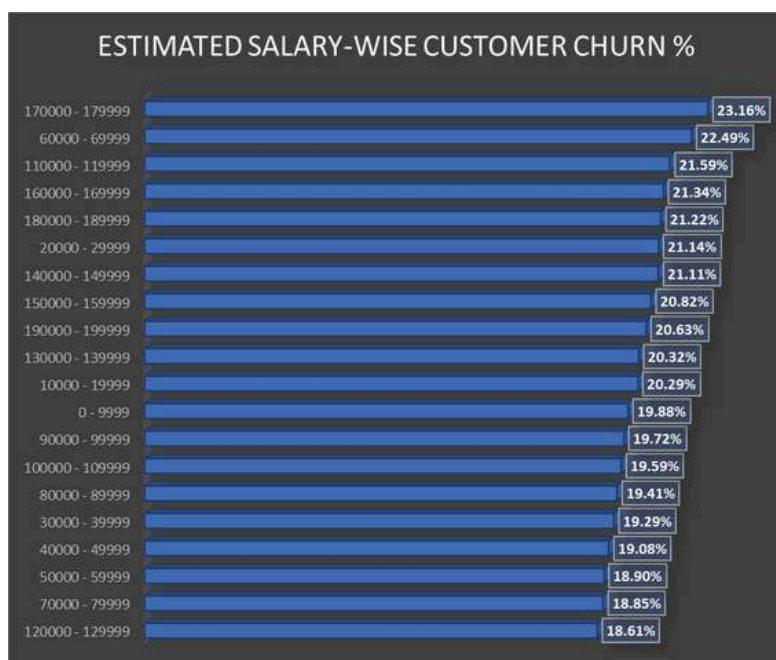
## 7. Customer Churn by Estimated Salary

To analyze whether or not customer income level influences churn behavior, customers were grouped into estimated salary ranges of ₹10,000 intervals and evaluated based on churn distribution.

SQL Query Used:

```
SELECT
 CONCAT(FLOOR(estimatedsalary / 10000) * 10000,
 '-',
 FLOOR(estimatedsalary / 10000) * 10000 + 9999) AS estimatedsalary_range,
 COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
 COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count,
 COUNT(*) AS total_customers,
 ROUND(
 (COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2)
 AS churn_percentage FROM bank_churn WHERE estimatedsalary <= 300000 GROUP BY FLOOR(estimatedsalary / 10000)
ORDER BY churn_percentage DESC;
```

|    | estimated_salary_range<br>text | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|--------------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | 170000 - 179999                | 123                    | 408                      | 531                       | 23.16                       |
| 2  | 60000 - 69999                  | 114                    | 393                      | 507                       | 22.49                       |
| 3  | 110000 - 119999                | 106                    | 385                      | 491                       | 21.59                       |
| 4  | 160000 - 169999                | 102                    | 376                      | 478                       | 21.34                       |
| 5  | 180000 - 189999                | 101                    | 375                      | 476                       | 21.22                       |
| 6  | 20000 - 29999                  | 104                    | 388                      | 492                       | 21.14                       |
| 7  | 140000 - 149999                | 110                    | 411                      | 521                       | 21.11                       |
| 8  | 150000 - 159999                | 96                     | 365                      | 461                       | 20.82                       |
| 9  | 190000 - 199999                | 105                    | 404                      | 509                       | 20.63                       |
| 10 | 130000 - 139999                | 102                    | 400                      | 502                       | 20.32                       |
| 11 | 10000 - 19999                  | 97                     | 381                      | 478                       | 20.29                       |
| 12 | 0 - 9999                       | 101                    | 407                      | 508                       | 19.88                       |
| 13 | 90000 - 99999                  | 97                     | 395                      | 492                       | 19.72                       |
| 14 | 100000 - 109999                | 105                    | 431                      | 536                       | 19.59                       |
| 15 | 80000 - 89999                  | 99                     | 411                      | 510                       | 19.41                       |
| 16 | 30000 - 39999                  | 92                     | 385                      | 477                       | 19.29                       |
| 17 | 40000 - 49999                  | 95                     | 403                      | 498                       | 19.08                       |
| 18 | 50000 - 59999                  | 96                     | 412                      | 508                       | 18.90                       |
| 19 | 70000 - 79999                  | 98                     | 422                      | 520                       | 18.85                       |
| 20 | 120000 - 129999                | 94                     | 411                      | 505                       | 18.61                       |



A bar chart was used to compare churn percentages across estimated salary ranges.

Additional comparison was made between exited and total customer counts for each group



### Insight

The analysis shows that churn percentages remain relatively stable across most estimated salary ranges, generally fluctuating between **18% and 23%**.

The highest churn rate is observed in the **₹170,000–₹179,999 salary range (23.16%)**, while the lowest appears in the **₹120,000–₹129,999 range (18.61%)**. However, the variation across salary groups is relatively small and does not indicate a strong linear relationship between salary and customer churn.

Additionally, both exited and retained customers are distributed fairly consistently across all income ranges.

### Business Interpretation

Estimated salary does not appear to be a strong standalone predictor of churn in this dataset. The relatively uniform churn percentages across salary segments suggest that customer exit behavior is not significantly influenced by income level alone. Customers across both lower and higher salary brackets demonstrate comparable retention patterns.

This indicates that:

- financial income may not directly determine customer satisfaction or loyalty,
- and other behavioral or engagement-related variables are likely to have greater impact on churn behavior.

## Key Takeaway

Estimated salary is a **weak-to-moderate churn indicator** and should not be treated as a primary metric for churn prediction.

While slight variations exist between salary groups, the overall churn distribution remains relatively consistent. Therefore, estimated salary is more useful as a **supporting contextual variable** when combined with stronger churn factors such as age, geography, product usage, and customer activity status.

## 8. Customer Churn by Tenure

To evaluate whether or not customer relationship duration influences churn behavior, customers were segmented based on tenure (number of years associated with the bank) and analyzed using churn percentages.

SQL Query Used:

```
SELECT tenure, COUNT(*) AS total_customers, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count,
ROUND(((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage
FROM bank_churn GROUP BY tenure ORDER BY churn_percentage DESC;
```

|    | tenure<br>integer | total_customers<br>bigint | exited_count<br>bigint | retained_count<br>bigint | churn_percentage<br>numeric |
|----|-------------------|---------------------------|------------------------|--------------------------|-----------------------------|
| 1  | 0                 | 413                       | 95                     | 318                      | 23.00                       |
| 2  | 1                 | 1035                      | 232                    | 803                      | 22.42                       |
| 3  | 9                 | 984                       | 213                    | 771                      | 21.65                       |
| 4  | 3                 | 1009                      | 213                    | 796                      | 21.11                       |
| 5  | 5                 | 1012                      | 209                    | 803                      | 20.65                       |
| 6  | 10                | 490                       | 101                    | 389                      | 20.61                       |
| 7  | 4                 | 989                       | 203                    | 786                      | 20.53                       |
| 8  | 6                 | 967                       | 196                    | 771                      | 20.27                       |
| 9  | 8                 | 1025                      | 197                    | 828                      | 19.22                       |
| 10 | 2                 | 1048                      | 201                    | 847                      | 19.18                       |
| 11 | 7                 | 1028                      | 177                    | 851                      | 17.22                       |



A bar chart was used to compare churn percentages across tenure.



Additional comparison was made between exited and total customer counts for each group

### Insight

The analysis shows that churn percentages remain relatively consistent across most tenure groups, ranging between approximately **17% and 23%**.

Customers with:

- **0 years tenure** show the highest churn rate at **23.00%**
- while customers with **7 years tenure** exhibit the lowest churn rate at **17.22%**

However, the variation across tenure categories is relatively moderate, and no clear increasing or decreasing trend is observed as tenure increases.

Additionally, exited and retained customer counts remain proportionally distributed across nearly all tenure levels.

### Business Interpretation

The findings suggest that tenure has a limited influence on customer churn behavior.

Although newly acquired customers (0–1 years) show slightly higher churn rates, long-term customers also continue to churn at comparable levels. This indicates that simply maintaining a longer banking relationship does not necessarily guarantee customer retention.

The absence of a strong trend suggests that churn is likely influenced more by:

- customer experience,
- service quality,
- engagement levels,
- and financial behavior,

rather than relationship duration alone.

### Key Takeaway

Tenure should be considered a **weak churn indicator** in this dataset.

Since churn percentages remain relatively stable across most tenure groups, tenure alone is not sufficient for accurately predicting customer exit behavior. It may still provide supporting context when combined with stronger variables such as age, geography, balance, and product usage.

## 9. Customer Churn by Number of Products 🏠

To examine the relationship between product usage and customer retention, customers were segmented based on the number of banking products they hold with the bank.

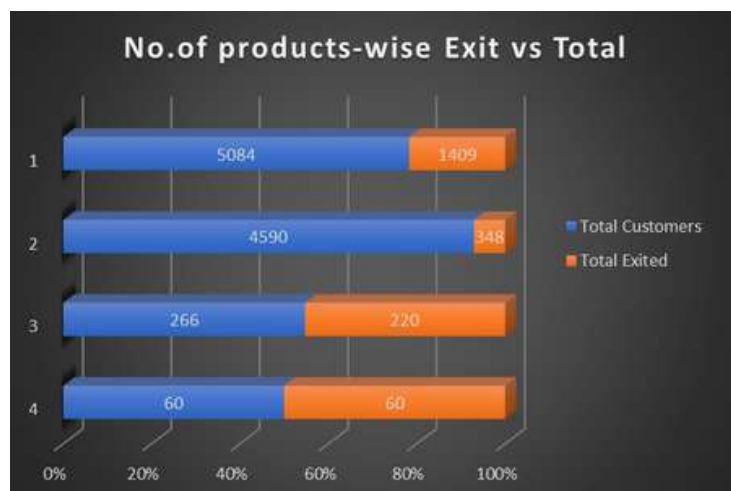
SQL Query Used:

```
SELECT
 numofproducts,
 COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
 COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count,
 COUNT(*) AS total_customers,
 ROUND(
 (COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2
) AS churn_percentage
FROM bank_churn
GROUP BY numofproducts
ORDER BY churn_percentage DESC;
```

|   | numofproducts<br>Integer | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|---|--------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1 | 4                        | 60                     | 0                        | 60                        | 100.00                      |
| 2 | 3                        | 220                    | 46                       | 266                       | 82.71                       |
| 3 | 1                        | 1409                   | 3675                     | 5084                      | 27.71                       |
| 4 | 2                        | 348                    | 4242                     | 4590                      | 7.58                        |



A pie chart was used to compare churn percentages across product categories.



Additional comparison was made between exited and total customer counts for each group

### Insight

The analysis reveals a **very strong relationship** between the number of products held and customer churn behavior.

Customers with:

- **2 products** show the lowest churn rate at **7.58%**
- while customers with **3 products** exhibit an extremely high churn rate of **82.71%**
- customers with **4 products** show a churn rate of **100%**, although this group contains only 60 customers

Additionally, customers holding only **1 product** demonstrate a relatively high churn rate of **27.71%**. The contrast between customers with 1 product and 2 products is particularly significant.

### Business Interpretation

The findings suggest that **product engagement plays a major role in customer retention**.

Customers using multiple banking services are generally more integrated into the bank's ecosystem, making them less likely to leave. The exceptionally low churn among customers with 2 products indicates stronger engagement and higher customer stickiness.

However, the unusually high churn rates among customers with 3 and 4 products may indicate:

- potential dissatisfaction among high-engagement customers,
- product complexity issues,
- or possible data imbalance due to smaller sample sizes.

Since the 3-product and 4-product segments contain relatively few customers compared to other categories, these results should be interpreted cautiously.

## Key Takeaway

The **number of products is one of the strongest churn indicators** identified in this analysis. The dataset clearly demonstrates that customer product engagement significantly influences retention behavior, making product usage an important metric for churn prediction and customer retention strategy development.

### 10. Customer Churn by Credit Card Ownership 🇮🇳

To determine whether or not credit card ownership influences customer retention behavior, customers were segmented based on whether they hold a credit card with the bank.

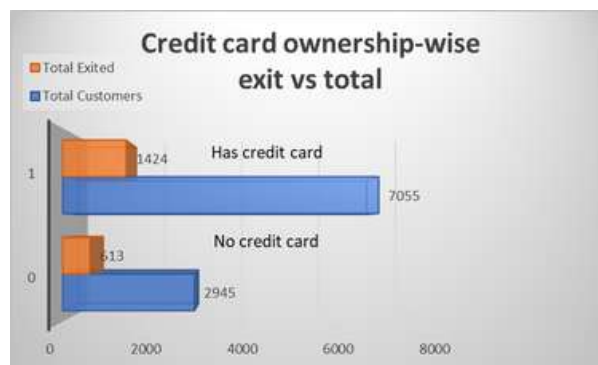
SQL Query Used:

```
SELECT hasccard, COUNT(*) AS total_customers, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count,
ROUND(
 (COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2
) AS churn_percentage FROM bank_churn GROUP BY hasccard ORDER BY churn_percentage DESC;
```

|   | hasccard<br>boolean | total_customers<br>bigint | exited_count<br>bigint | retained_count<br>bigint | churn_percentage<br>numeric |
|---|---------------------|---------------------------|------------------------|--------------------------|-----------------------------|
| 1 | false               | 2945                      | 613                    | 2332                     | 20.81                       |
| 2 | true                | 7055                      | 1424                   | 5631                     | 20.18                       |



A pie chart was used to compare churn percentages across credit card ownership.



Additional comparison was made between exited and total customer counts for each group

## Insight

The analysis shows very little difference in churn behavior between customers who own a credit card and those who do not.

- Customers without credit cards show a churn rate of **20.81%**
- Customers with credit cards show a churn rate of **20.18%**

The variation between the two groups is less than 1%, indicating a highly similar churn distribution.

## Business Interpretation

Credit card ownership does not appear to significantly influence customer retention in this dataset. Although a larger number of customers possess credit cards, the churn percentages remain nearly identical across both groups. This suggests that simply offering or owning a credit card is not enough to meaningfully impact customer loyalty or reduce attrition.

The findings indicate that churn behavior is likely driven more strongly by customer engagement, financial behavior, geography, and product usage patterns, rather than credit card ownership alone.

## Key Takeaway

Credit card ownership is a **weak churn indicator** and does not provide strong predictive value for customer exit behavior in this dataset.

Due to the minimal variation in churn percentages between both groups, this variable **should not be considered a primary metric for churn analysis**.

### 11. Customer Churn by Active Membership Status

To evaluate the impact of customer engagement on churn behavior, customers were segmented based on their active membership status.

SQL Query Used:

```
SELECT isactivemember, COUNT(*) AS total_customers, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count,
ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage
FROM bank_churn GROUP BY isactivemember ORDER BY churn_percentage DESC;
```

|   | isactivemember<br>boolean | total_customers<br>bigint | exited_count<br>bigint | retained_count<br>bigint | churn_percentage<br>numeric |
|---|---------------------------|---------------------------|------------------------|--------------------------|-----------------------------|
| 1 | false                     | 4849                      | 1302                   | 3547                     | 26.85                       |
| 2 | true                      | 5151                      | 735                    | 4416                     | 14.27                       |

### ACTIVE MEMBERSHIP-WISE CUSTOMER CHURN %



A pie chart was used to compare churn percentages between active and inactive customers



Additional comparison was made between exited and total customer counts for each group

## Insight

The analysis reveals a substantial difference in churn behavior based on customer activity status.

**Inactive customers exhibit a significantly higher churn rate of 26.85%, compared to only 14.27% among active members.**

This indicates that inactive customers are nearly twice as likely to discontinue banking services compared to customers who actively engage with the bank.

## Business Interpretation

Customer engagement appears to play a major role in retention behavior.

Active members are likely:

- interacting more frequently with banking services,
- deriving greater value from the platform,
- and maintaining stronger engagement with the bank's ecosystem.

In contrast, inactive customers may:

- experience lower product attachment,
- reduced service usage,
- or weaker overall engagement,
- making them more vulnerable to churn.

### Key Takeaway

Active membership status is a **strong churn indicator** in this dataset.

The significant difference in churn percentages between active and inactive customers highlights the importance of customer engagement and activity tracking in churn prediction and retention strategy development.

Improving customer engagement may therefore be an effective approach for reducing customer attrition.

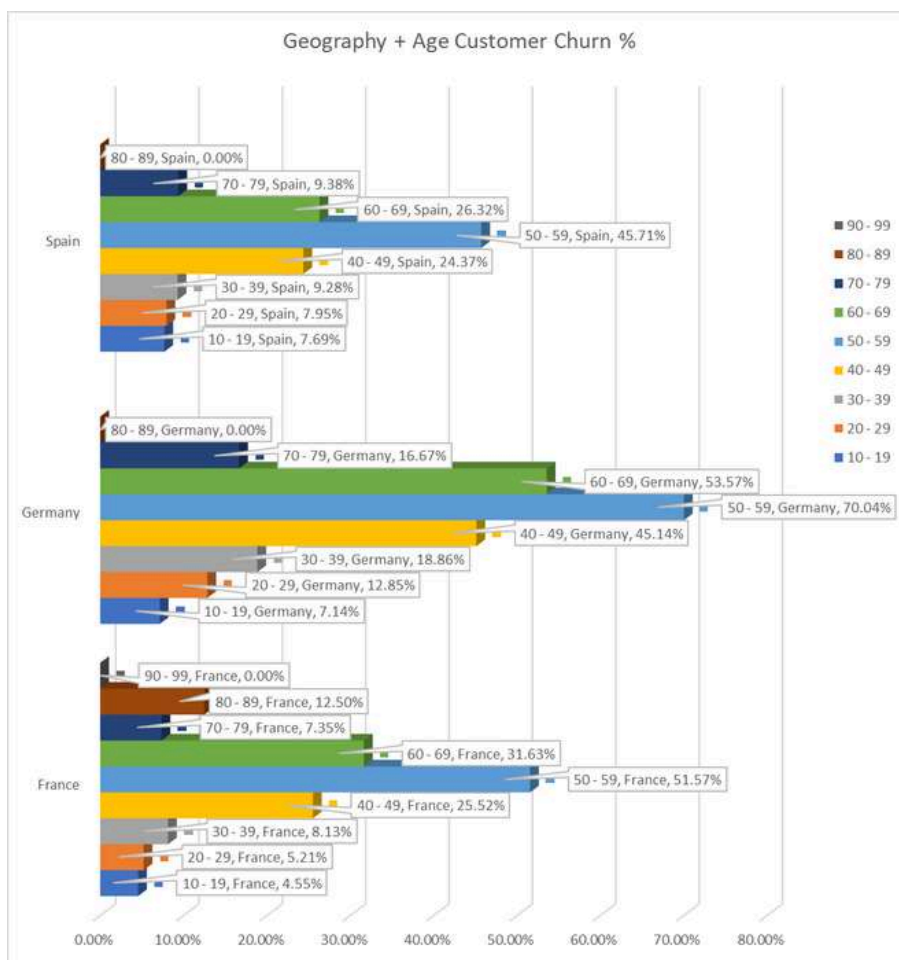
## LEVEL 2: Combined Attribute Analysis

### 1. Geography and Age Combined Analysis 🌍👤

To identify high-risk customer segments across different regions, customer churn was analyzed using a combined segmentation of geography and age groups.

SQL Query Used:

```
SELECT geography, CONCAT(FLOOR(age/10) * 10, ' - ', FLOOR(age/10) * 10 + 9) AS age_range, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn WHERE age <= 100 GROUP BY floor(age/10), geography ORDER BY churn_percentage desc;
```



## Key Observations

The analysis revealed that Germany consistently exhibited the highest churn percentages across middle-aged customer groups.

| Geography | Age Range | Churn %       |
|-----------|-----------|---------------|
| Germany   | 50-59     | <b>70.04%</b> |
| Germany   | 60-69     | <b>53.57%</b> |
| France    | 50-59     | 51.57%        |
| Spain     | 50-59     | 45.71%        |
| Germany   | 40-49     | 45.14%        |

|    | geography<br>character_varying (20) | age_range<br>text | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|-------------------------------------|-------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | Germany                             | 50-59             | 194                    | 83                       | 277                       | 70.04                       |
| 2  | Germany                             | 60-69             | 45                     | 39                       | 84                        | 53.57                       |
| 3  | France                              | 50-59             | 197                    | 185                      | 382                       | 51.57                       |
| 4  | Spain                               | 50-59             | 96                     | 114                      | 210                       | 45.71                       |
| 5  | Germany                             | 40-49             | 334                    | 405                      | 740                       | 45.14                       |
| 6  | France                              | 60-69             | 62                     | 134                      | 196                       | 31.63                       |
| 7  | Spain                               | 60-69             | 25                     | 70                       | 95                        | 26.32                       |
| 8  | France                              | 40-49             | 317                    | 925                      | 1242                      | 25.52                       |
| 9  | Spain                               | 40-49             | 155                    | 481                      | 636                       | 34.37                       |
| 10 | Germany                             | 30-39             | 188                    | 809                      | 997                       | 18.86                       |
| 11 | Germany                             | 70-79             | 5                      | 30                       | 35                        | 16.67                       |
| 12 | Germany                             | 20-29             | 46                     | 312                      | 358                       | 12.85                       |
| 13 | France                              | 80-89             | 1                      | 7                        | 8                         | 12.50                       |
| 14 | Spain                               | 70-79             | 3                      | 29                       | 32                        | 9.38                        |
| 15 | Spain                               | 30-39             | 102                    | 997                      | 1099                      | 9.28                        |
| 16 | France                              | 30-39             | 183                    | 2067                     | 2250                      | 8.13                        |
| 17 | Spain                               | 20-29             | 31                     | 359                      | 390                       | 7.95                        |
| 18 | Spain                               | 10-19             | 1                      | 12                       | 13                        | 7.69                        |
| 19 | France                              | 70-79             | 5                      | 63                       | 68                        | 7.35                        |
| 20 | Germany                             | 10-19             | 1                      | 13                       | 14                        | 7.14                        |
| 21 | France                              | 20-29             | 44                     | 800                      | 844                       | 5.21                        |
| 22 | France                              | 10-19             | 1                      | 21                       | 22                        | 4.55                        |
| 23 | Germany                             | 80-89             | 0                      | 3                        | 3                         | 0.00                        |
| 24 | Spain                               | 80-89             | 0                      | 2                        | 2                         | 0.00                        |
| 25 | France                              | 90-99             | 0                      | 2                        | 2                         | 0.00                        |

Additionally:

- Customers aged **40-69** showed **significantly higher churn rates across all three countries**
- Younger customer groups (20-39) generally demonstrated lower churn percentages
- Germany emerged as the highest-risk geography across nearly all major age categories

## Insight

The combined analysis suggests that both geography and age strongly influence churn behavior when analyzed together.

Middle-aged customers, particularly those in Germany, appear substantially more likely to discontinue banking services compared to younger customers and customers from other regions. This indicates that customer churn risk is concentrated within specific demographic and regional segments rather than being evenly distributed across the customer base.

## Business Interpretation

The findings highlight the importance of targeted retention strategies for middle-aged customers in high-risk geographic regions such as Germany.

Possible contributing factors may include:

- increased competition within regional banking markets,
- changing financial priorities among older customer groups,
- or differences in customer expectations and banking engagement.

## Key Takeaway

The combination of geography and age emerged as one of the strongest multi-variable churn indicators in the dataset.

Customers aged 40-69 in Germany represent a particularly high-risk churn segment and may benefit from focused retention initiatives and personalized engagement strategies.

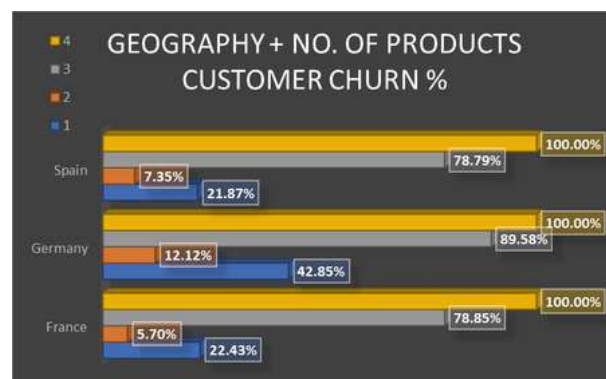
## 2. Geography and Number of Products Combined Analysis 🏠🌐

To understand how product usage influences churn across different regions, customer churn was analyzed using a combined segmentation of geography and number of banking products held.

SQL Query Used:

```
select geography, numofproducts, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn group by numofproducts, geography order by churn_percentage desc;
```

|    | geography<br>character varying (20) | numofproducts<br>integer | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|-------------------------------------|--------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | France                              | 4                        | 29                     | 0                        | 29                        | 100.00                      |
| 2  | Germany                             | 4                        | 24                     | 0                        | 24                        | 100.00                      |
| 3  | Spain                               | 4                        | 7                      | 0                        | 7                         | 100.00                      |
| 4  | Germany                             | 3                        | 86                     | 10                       | 96                        | 89.58                       |
| 5  | France                              | 3                        | 82                     | 22                       | 104                       | 78.85                       |
| 6  | Spain                               | 3                        | 52                     | 14                       | 66                        | 78.79                       |
| 7  | Germany                             | 1                        | 578                    | 771                      | 1349                      | 42.85                       |
| 8  | France                              | 1                        | 564                    | 1950                     | 2514                      | 22.43                       |
| 9  | Spain                               | 1                        | 267                    | 954                      | 1221                      | 21.87                       |
| 10 | Germany                             | 2                        | 126                    | 914                      | 1040                      | 12.12                       |
| 11 | Spain                               | 2                        | 87                     | 1096                     | 1183                      | 7.35                        |
| 12 | France                              | 2                        | 135                    | 2232                     | 2367                      | 5.70                        |



### Key Observations

The analysis shows a very strong relationship between product usage and churn, with clear differences across geographic regions.

- Customers holding **3–4 products** across all geographies show **extremely high churn rates (78%–100%)**, although these segments **contain relatively small sample sizes**.
- Customers with **1 product show moderate to high churn**, especially in **Germany (42.85%)**, compared to France (22.43%) and Spain (21.87%).
- Customers with **2 products consistently show the lowest churn rates across all countries**, particularly in France (5.70%) and Spain (7.35%).

### Insight

The combined analysis indicates that number of products is one of the strongest determinants of customer churn across all geographies.

Customers with only one product or very high product counts (3–4) show significantly higher churn rates, while customers with 2 products demonstrate the highest retention levels across all regions.

### Business Interpretation

The findings suggest that customer engagement through product ownership plays a critical role in retention, regardless of geography.

- Customers with **multiple (but balanced) product usage (2 products) are the most stable segment**.
- Over-engagement (3–4 products) may indicate complexity or dissatisfaction.
- Low engagement (1 product) reflects weak customer integration with the bank.

This pattern is consistent across France, Spain, and Germany, indicating a universal behavioral trend rather than region-specific effects.

### Key Takeaway

Geography does not significantly alter the strong relationship between product usage and churn behavior.

The number of products remains a dominant churn driver across all regions, with customers holding 2 products showing the strongest retention patterns in every country.

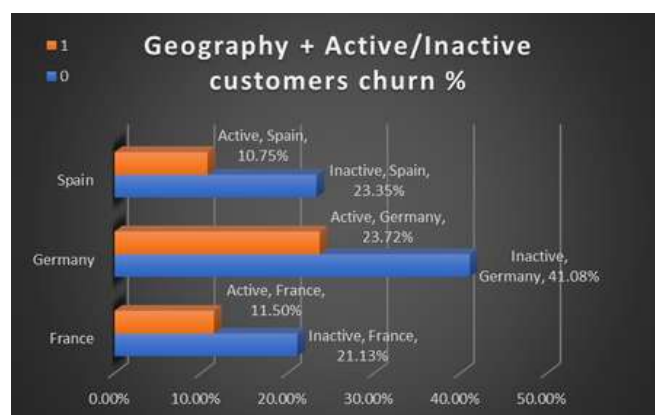
### 3. Geography and Active Membership Combined Analysis 📌

To evaluate the combined impact of geography and customer engagement, churn was analyzed across regions based on active membership status.

SQL Query Used:

```
SELECT geography, isactivemember, COUNT(*) AS total_customers, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 /
COUNT(*)), 2) AS churn_percentage FROM bank_churn GROUP BY isactivemember, geography ORDER BY churn_percentage DESC;
```

|   | geography<br>character varying (20) | isactivemember<br>boolean | total_customers<br>bigint | exited_count<br>bigint | retained_count<br>bigint | churn_percentage<br>numeric |
|---|-------------------------------------|---------------------------|---------------------------|------------------------|--------------------------|-----------------------------|
| 1 | Germany                             | false                     | 1261                      | 518                    | 743                      | 41.08                       |
| 2 | Germany                             | true                      | 1248                      | 296                    | 952                      | 23.72                       |
| 3 | Spain                               | false                     | 1165                      | 272                    | 893                      | 23.35                       |
| 4 | France                              | false                     | 2423                      | 512                    | 1911                     | 21.13                       |
| 5 | France                              | true                      | 2591                      | 298                    | 2293                     | 11.50                       |
| 6 | Spain                               | true                      | 1312                      | 141                    | 1171                     | 10.75                       |



#### Key Observations

The analysis shows a clear and consistent pattern across all geographies:

- **Inactive customers in Germany** show the **highest churn rate at 41.08%**, significantly higher than **active German customers (23.72%)**.
- In **France**, **inactive** customers also exhibit higher churn (**21.13%**) compared to **active** members (**11.50%**).
- A similar trend is observed in **Spain**, where **inactive** customers churn at **23.35%**, while **active** customers show a much lower churn rate of **10.75%**.

Across all regions, **inactive members** consistently demonstrate nearly **2x higher churn rates** than active members.

#### Insight

Active membership status is a strong and consistent driver of customer retention across all geographies.

While geography influences baseline churn levels (with Germany showing higher overall churn), customer activity status significantly amplifies or reduces churn risk within each region.

#### Business Interpretation

The findings highlight **customer engagement as a critical factor in retention strategy**.

Inactive customers are significantly more likely to churn, regardless of geography, indicating that **lack of engagement is a stronger churn driver** than regional differences alone.

This suggests that improving customer activity levels—through better product engagement, usage incentives, or personalized interactions—could have a direct impact on reducing churn across all markets.

#### Key Takeaway

**Active membership status is a highly influential churn factor across all geographies.**

Customers who are inactive consistently show significantly higher churn rates, making engagement-driven retention strategies essential for reducing attrition.

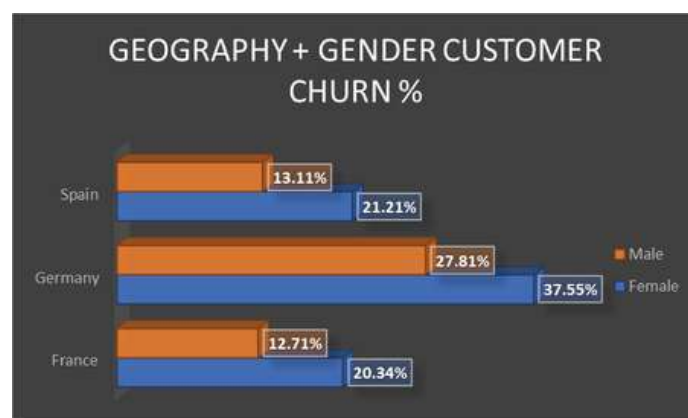
#### 4. Geography and Gender Combined Analysis 🧑🌍

To understand how churn varies across gender within different geographic regions, customer churn was analyzed using a combined segmentation of geography and gender.

SQL Query Used:

```
select geography, gender, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE)
AS retained_count, count (*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2)
AS churn_percentage FROM bank_churn GROUP BY gender, geography ORDER BY churn_percentage desc;
```

|   | geography<br>character varying (20) | gender<br>character varying (10) | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|---|-------------------------------------|----------------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1 | Germany                             | Female                           | 448                    | 745                      | 1193                      | 37.55                       |
| 2 | Germany                             | Male                             | 366                    | 950                      | 1316                      | 27.81                       |
| 3 | Spain                               | Female                           | 231                    | 858                      | 1089                      | 21.21                       |
| 4 | France                              | Female                           | 460                    | 1801                     | 2261                      | 20.34                       |
| 5 | Spain                               | Male                             | 182                    | 1206                     | 1388                      | 13.11                       |
| 6 | France                              | Male                             | 350                    | 2403                     | 2753                      | 12.71                       |



#### Key Observations

The analysis reveals clear differences in churn behavior across both geography and gender:

- **Female customers in Germany** show the **highest churn rate at 37.55%**, significantly higher than German male customers (27.81%).
- In **Spain**, **female customers** also exhibit higher churn (**21.21%**) compared to males (13.11%).
- In **France**, **female churn (20.34%)** is again higher than male churn (12.71%).

Across all three countries, female customers consistently demonstrate higher churn rates than male customers.

#### Insight

Gender-based differences in churn are consistent across all geographies, with female customers showing higher attrition rates regardless of region.

However, geography still plays a role in overall churn levels, with Germany showing the highest churn across both genders.

#### Business Interpretation

The findings suggest that churn behavior is influenced by an interaction of demographic and regional factors.

While geography determines baseline churn levels, gender further differentiates risk within each region, with **female customers representing a higher-risk segment across all countries**.

This indicates that retention strategies may need to consider both regional and gender-based segmentation to effectively reduce churn.

#### Key Takeaway

Female customers consistently show higher churn rates across all geographies.

This makes gender a meaningful secondary segmentation factor when combined with geography, helping identify higher-risk customer groups within each region.

## Other Combined Analysis:

### 5. Gender + Credit Score Analysis

Female customers consistently showed higher churn percentages than male customers across nearly all credit score ranges.

For example:

- **Female** customers in the **600–699** score range showed a churn rate of **24.61%**, compared to **15.61%** for **male** customers.
- Similarly, in the **700–799** range, **female** churn was **25.09%** versus **15.52%** for **males**.

This suggests that gender differences in churn behavior persist regardless of customer credit score category.

|    | gender<br>character varying (10) | creditscore_range<br>text | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|----------------------------------|---------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | Female                           | 300 - 399                 | 11                     | 0                        | 11                        | 100.00                      |
| 2  | Male                             | 300 - 399                 | 8                      | 0                        | 8                         | 100.00                      |
| 3  | Female                           | 400 - 499                 | 73                     | 212                      | 285                       | 25.61                       |
| 4  | Female                           | 500 - 599                 | 271                    | 799                      | 1070                      | 25.33                       |
| 5  | Female                           | 700 - 799                 | 286                    | 854                      | 1140                      | 25.09                       |
| 6  | Female                           | 600 - 699                 | 429                    | 1314                     | 1743                      | 24.61                       |
| 7  | Female                           | 800 - 899                 | 69                     | 225                      | 294                       | 23.47                       |
| 8  | Male                             | 500 - 599                 | 239                    | 1093                     | 1332                      | 17.94                       |
| 9  | Male                             | 400 - 499                 | 58                     | 270                      | 328                       | 17.68                       |
| 10 | Male                             | 800 - 899                 | 59                     | 302                      | 361                       | 16.34                       |
| 11 | Male                             | 600 - 699                 | 324                    | 1751                     | 2075                      | 15.61                       |
| 12 | Male                             | 700 - 799                 | 210                    | 1143                     | 1353                      | 15.52                       |

### 6. Gender + Age Analysis

Customer churn was further analyzed by combining gender and age groups to identify demographic segments with higher churn tendencies.

The analysis revealed that female customers consistently showed higher churn percentages than male customers across most age categories.

Key observations include:

- **Female** customers aged **50–59** showed the highest churn rate at **62.95%**, compared to **49.55%** for **males** in the same age group.
- Female customers in the 40–49 age range exhibited a churn rate of 36.09%, and female customers in the 60–69 range exhibited a churn rate of 43.03%, while male customers in the same categories showed 25.97% & 29.05% churn respectively.
- Younger customers (20–39 age groups) generally demonstrated lower churn percentages across both genders.

These findings suggest that churn risk increases significantly among middle-aged customers, particularly female customers between the ages of 40 and 69.

|    | gender<br>character varying (10) | age_range<br>text | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|----------------------------------|-------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | Female                           | 50 - 59           | 265                    | 156                      | 421                       | 62.95                       |
| 2  | Male                             | 50 - 59           | 222                    | 226                      | 448                       | 49.55                       |
| 3  | Female                           | 60 - 69           | 71                     | 94                       | 165                       | 43.03                       |
| 4  | Female                           | 40 - 49           | 450                    | 797                      | 1247                      | 36.09                       |
| 5  | Male                             | 60 - 69           | 61                     | 149                      | 210                       | 29.05                       |
| 6  | Male                             | 40 - 49           | 356                    | 1015                     | 1371                      | 25.97                       |
| 7  | Female                           | 80 - 89           | 1                      | 5                        | 6                         | 16.67                       |
| 8  | Female                           | 70 - 79           | 10                     | 60                       | 70                        | 14.29                       |
| 9  | Female                           | 30 - 39           | 268                    | 1616                     | 1884                      | 14.23                       |
| 10 | Female                           | 20 - 29           | 73                     | 664                      | 737                       | 9.91                        |
| 11 | Male                             | 30 - 39           | 205                    | 2257                     | 2462                      | 8.33                        |
| 12 | Female                           | 10 - 19           | 1                      | 12                       | 13                        | 7.69                        |
| 13 | Male                             | 70 - 79           | 4                      | 62                       | 66                        | 6.06                        |
| 14 | Male                             | 20 - 29           | 48                     | 807                      | 855                       | 5.61                        |
| 15 | Male                             | 10 - 19           | 2                      | 34                       | 36                        | 5.56                        |
| 16 | Male                             | 90 - 99           | 0                      | 2                        | 2                         | 0.00                        |
| 17 | Male                             | 80 - 89           | 0                      | 7                        | 7                         | 0.00                        |

## 7. Geography + Credit Score Analysis

Customer churn was further analyzed across combined geography and credit score segments to identify regional differences in financial-profile-based churn behavior.

The analysis showed that Germany consistently exhibited higher churn percentages across nearly all major credit score ranges compared to France and Spain.

Key observations include:

- **German** customers in the **500–599 credit score range** showed a churn rate of **34.52%**
- German customers in the **700–799 range** exhibited **33.65%** churn
- In comparison, **France and Spain** generally maintained churn percentages **between 14% and 18%** across similar credit score categories

These findings suggest that **geographic location has a stronger influence on churn behavior than credit score alone**, with **Germany emerging as the highest-risk region** across multiple financial segments.

|    | geography<br>character varying (20) | creditscore_range<br>text | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|-------------------------------------|---------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | Germany                             | 300 - 399                 | 6                      | 0                        | 6                         | 100.00                      |
| 2  | Spain                               | 300 - 399                 | 7                      | 0                        | 7                         | 100.00                      |
| 3  | France                              | 300 - 399                 | 6                      | 0                        | 6                         | 100.00                      |
| 4  | Germany                             | 500 - 599                 | 203                    | 385                      | 588                       | 34.52                       |
| 5  | Germany                             | 700 - 799                 | 213                    | 420                      | 633                       | 33.65                       |
| 6  | Germany                             | 400 - 499                 | 55                     | 111                      | 166                       | 33.13                       |
| 7  | Germany                             | 600 - 699                 | 285                    | 655                      | 940                       | 30.32                       |
| 8  | Germany                             | 800 - 899                 | 52                     | 124                      | 176                       | 29.55                       |
| 9  | France                              | 400 - 499                 | 58                     | 263                      | 321                       | 18.07                       |
| 10 | Spain                               | 500 - 599                 | 105                    | 505                      | 610                       | 17.21                       |
| 11 | France                              | 500 - 599                 | 202                    | 1002                     | 1204                      | 16.78                       |
| 12 | Spain                               | 700 - 799                 | 103                    | 512                      | 615                       | 16.75                       |
| 13 | Spain                               | 800 - 899                 | 26                     | 130                      | 156                       | 16.67                       |
| 14 | France                              | 600 - 699                 | 314                    | 1601                     | 1915                      | 16.40                       |
| 15 | Spain                               | 600 - 699                 | 154                    | 809                      | 963                       | 15.99                       |
| 16 | France                              | 800 - 899                 | 50                     | 273                      | 323                       | 15.48                       |
| 17 | France                              | 700 - 799                 | 180                    | 1065                     | 1245                      | 14.46                       |
| 18 | Spain                               | 400 - 499                 | 18                     | 108                      | 126                       | 14.29                       |

## B) Summary Statistics

To understand the distribution and variability of key numerical variables, summary statistics including mean, median, mode, minimum, maximum, and standard deviation were computed.

The analysis focused on key financial and behavioral variables: Credit Score, Balance, Estimated Salary, Age, Tenure, and Number of Products.

### Key Observations

#### 1. Estimated Salary

- Mean: ~100,090
- Median: ~100,194
- Very close mean and median indicate a **nearly symmetric distribution**
- High standard deviation (~57,510) suggests wide variation across customers
- Range is extremely large (11.58 to 199,992)

**Insight:** Salary is widely spread but not heavily skewed.

|   | mean_salary<br>numeric | median_salary<br>double precision | mode_salary<br>numeric | min_salary<br>numeric | max_salary<br>numeric | stddev_salary<br>numeric |
|---|------------------------|-----------------------------------|------------------------|-----------------------|-----------------------|--------------------------|
| 1 | 100090.239881000000    | 100193.915                        | 24924.92               | 11.58                 | 199992.48             | 57510.49281770           |

#### 2. Balance

- Mean: ~76,486
- Median: ~97,199
- Median is significantly higher than mean
- Mode is 0 (many customers have zero balance)
- High standard deviation (~62,397)

**Insight:** Strong **left-skew due to many zero-balance customers**, indicating presence of a large inactive/low-usage segment.

|   | mean_balance<br>numeric | median_balance<br>double precision | mode_balance<br>numeric | min_balance<br>numeric | max_balance<br>numeric | stddev_balance<br>numeric |
|---|-------------------------|------------------------------------|-------------------------|------------------------|------------------------|---------------------------|
| 1 | 76485.889288000000      | 97198.54000000001                  | 0                       | 0                      | 250898.09              | 62397.40520239            |

#### 3. Credit Score

- Mean: ~650
- Median: 652
- Relatively symmetric distribution
- Range: 350 to 850
- Moderate standard deviation (~96)

**Insight:** Credit scores are evenly distributed with no major skewness.

|   | mean_creditscore<br>numeric | median_creditscore<br>double precision | mode_creditscore<br>integer | min_creditscore<br>integer | max_creditscore<br>integer | stddev_creditscore<br>numeric |
|---|-----------------------------|----------------------------------------|-----------------------------|----------------------------|----------------------------|-------------------------------|
| 1 | 650.5288000000000000        | 652                                    | 850                         | 350                        | 850                        | 96.6532987361303544           |

#### 4. Age

- Mean: ~39
- Median: 37
- Slight right skew (older outliers exist up to 92)
- Moderate variation (std ~10.5)

**Insight:** Majority of customers are middle-aged with a small elderly segment.

|   | mean_age<br>numeric | median_age<br>double precision | mode_age<br>integer | min_age<br>integer | max_age<br>integer | stddev_age<br>numeric |
|---|---------------------|--------------------------------|---------------------|--------------------|--------------------|-----------------------|
| 1 | 38.921800000000000  | 37                             | 37                  | 18                 | 92                 | 10.4878064517046100   |

#### 5. Tenure

- Mean  $\approx$  Median  $\approx$  5
- Very stable distribution
- Low variation (std ~2.89)

**Insight:** Customer tenure is evenly distributed; not a strong differentiator.

|   | mean_tenure<br>numeric | median_tenure<br>double precision | mode_tenure<br>integer | min_tenure<br>integer | max_tenure<br>integer | stddev_tenure<br>numeric |
|---|------------------------|-----------------------------------|------------------------|-----------------------|-----------------------|--------------------------|
| 1 | 5.012800000000000      | 5                                 | 2                      | 0                     | 10                    | 2.8921743770496838       |

#### 6. Number of Products

- Mean: ~1.53
- Median: 1
- Mode: 1
- Low variation, but notable upper bound at 4 products
- Most customers hold only 1 product

**Insight:** Highly concentrated around low product usage.

|   | mean_numofproducts<br>numeric | median_numofproducts<br>double precision | mode_numofproducts<br>integer | min_numofproducts<br>integer | max_numofproducts<br>integer | stddev_numofproducts<br>numeric |
|---|-------------------------------|------------------------------------------|-------------------------------|------------------------------|------------------------------|---------------------------------|
| 1 | 1.5302000000000000            | 1                                        | 1                             | 1                            | 4                            | 0.58165435799899060138          |

### Key Analytical Insight

A comparison of mean and median values highlights important distribution patterns:

- Variables like **Balance** show noticeable skewness (**mean < median**), indicating **presence of outliers and a large group of zero-balance customers**.
- Variables like **Salary and Credit Score** show near-identical mean and median values, indicating **relatively balanced distributions**.
- Variables like **Age** show **mild skewness**, while **Tenure and Products** are **highly concentrated**, showing limited variability.

### Business Interpretation

The summary statistics confirm that:

- Financial variables (Balance, Salary) exhibit high dispersion.
- Behavioral variables (Products, Tenure) are more stable and clustered.
- Customer base is largely middle-aged with moderate credit scores.

This reinforces earlier churn findings by showing that variability is highest in financial behavior, while engagement patterns are more structured.

### Key Takeaway

Mean vs median comparison reveals that:

- Balance is the most skewed variable (outliers + zero-balance customers)
- Salary and Credit Score are well-distributed
- Product usage and tenure are highly concentrated and stable

This justifies why segmentation-based churn analysis is necessary instead of relying on averages alone.

## C) Key Churn Metrics Identified

Based on the SQL-based exploratory analysis, summary statistics, and combined attribute analysis, several variables emerged as significant indicators of customer churn behavior.

The metrics were categorized according to the strength and consistency of their relationship with churn patterns.

### ★ Strong Influencing Factors

#### 1. Age

Age emerged as one of the strongest churn indicators in the dataset.

Customers in the **40–69 age group** consistently showed significantly higher churn percentages compared to younger customers across all geographies and genders.

The strongest churn concentration was observed among customers aged 50–59, particularly in Germany.

#### **Why it matters:**

Age strongly influences customer financial behavior, engagement patterns, and banking expectations, making it a critical segmentation variable for churn analysis.

#### 2. Geography

Geographic location showed substantial differences in churn behavior.

Germany consistently demonstrated significantly higher churn rates (~32%) compared to France and Spain (~16%).

Combined attribute analysis further reinforced Germany as the highest-risk region across multiple customer segments.

#### **Why it matters:**

Regional market conditions, customer expectations, and banking competition may vary significantly across geographies, directly impacting customer retention.

#### 3. Number of Products

The number of products held by customers showed one of the clearest relationships with churn.

- Customers with **2 products** demonstrated the **lowest churn** rates.
- Customers with **1 product** or **3–4 products** showed **substantially higher churn** percentages.

#### **Why it matters:**

Product ownership reflects customer engagement and integration with the bank's services, making it a highly actionable retention metric.

#### 4. Active Membership Status

Active members consistently showed significantly lower churn rates compared to inactive customers across all geographies.

**Inactive customers in Germany exhibited churn rates exceeding 41%.**

**Why it matters:**

Customer engagement level is directly associated with retention behavior and serves as a strong indicator of customer loyalty.

#### ★ Moderate Influencing Factors

##### 5. Gender

Female customers consistently demonstrated higher churn percentages than male customers across nearly all major segments.

However, gender alone was not as strong a predictor as geography, age, or product usage.

**Why it matters:**

Gender contributes to churn variation when combined with other demographic and financial factors.

##### 6. Balance

Balance showed moderate influence on churn behavior.

Certain balance ranges exhibited elevated churn percentages, but the relationship was inconsistent due to large concentrations of customers with zero or moderate balances.

**Why it matters:**

Balance reflects customer financial engagement but is affected by skewed distribution and outliers.

##### 7. Credit Score

Credit score showed relatively stable churn percentages across most score ranges.

Although some regional differences existed, credit score alone did not strongly differentiate churn behavior.

**Why it matters:**

Credit score contributes supporting financial context but is not a dominant churn driver in this dataset.

#### ★ Weak / Non-Significant Factors

##### 8. Estimated Salary

Estimated salary demonstrated highly uniform churn percentages across salary ranges.

No major salary band showed significantly distinct churn behavior.

**Why it matters:**

Income level alone does not appear to strongly influence customer retention in this dataset.

##### 9. Tenure

Customer churn percentages remained relatively consistent across tenure categories.

Both churned and retained customers were distributed similarly across tenure values.

**Why it matters:**

Length of relationship with the bank does not independently explain churn behavior effectively.

## 10. Credit Card Ownership

Customers with and without credit cards showed nearly identical churn percentages (~20%).

### **Why it matters:**

Credit card ownership alone is not a meaningful differentiator of churn risk.

## RECOMMENDATIONS

Based on the findings from the churn analysis, the following business recommendations are proposed:

### 1. **Improve Engagement for Inactive Customers**

Inactive customers consistently exhibited significantly higher churn rates across all regions.

The company should:

- increase personalized engagement,
- encourage app usage,
- introduce loyalty programs,
- and provide proactive customer support initiatives.

### 2. **Strengthen Retention Efforts in Germany**

Germany showed the highest churn rates across nearly all customer segments.

The bank should:

- investigate region-specific customer dissatisfaction,
- improve localized customer experience,
- and implement targeted retention campaigns for German customers.

### 3. **Focus on Middle-Aged Customers**

Customers aged **40–69** demonstrated the highest churn tendencies.

Retention initiatives for this segment may include:

- personalized financial planning services,
- investment advisory offerings,
- and customized banking products.

### 4. **Encourage Balanced Product Adoption**

Customers with exactly **2 products** showed the strongest retention behavior.

The bank should encourage:

- cross-selling of complementary products,
- while avoiding customer over-complexity associated with excessive product ownership.

### 5. **Use Segmentation-Based Retention Strategies**

The analysis shows that churn is influenced by combinations of demographic, geographic, and behavioral variables rather than single metrics alone.

The bank should adopt:

- data-driven customer segmentation,
- predictive churn monitoring,
- and personalized retention strategies.

## CONCLUSION

This report analyzed customer churn patterns within the banking dataset using SQL-based exploratory analysis, summary statistics, and multi-variable segmentation techniques.

Out of 10,000 customers, **2,037 customers exited**, resulting in an overall **churn rate of 20.37%**.

The analysis identified:

- **Age, Geography, Number of Products, and Active Membership Status** as the strongest churn indicators,
- while variables such as Salary, Tenure, and Credit Card Ownership showed relatively weak influence on customer attrition.

Combined attribute analysis further revealed that churn risk is concentrated within specific customer segments, particularly:

- middle-aged customers,
- inactive members,
- customers in Germany,
- and customers with imbalanced product usage patterns.

The findings demonstrate the importance of data-driven customer segmentation in understanding churn behavior and designing effective retention strategies.

Overall, the report highlights how SQL-based business analytics can help financial institutions identify high-risk customer groups, improve retention initiatives, and support more informed business decision-making.

# APPENDIX

## SQL Analysis & Key Findings: Data Exploration - LEVEL 3 - Country-wise Analysis

### France

#### Age

|   | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|-------------------------------------|-----------------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | France                              | 50                                  | 59                                | 197                    | 185                    | 51.57                                               |
| 2 | France                              | 60                                  | 69                                | 62                     | 134                    | 31.63                                               |
| 3 | France                              | 40                                  | 49                                | 317                    | 925                    | 25.52                                               |
| 4 | France                              | 80                                  | 89                                | 1                      | 7                      | 12.50                                               |
| 5 | France                              | 30                                  | 39                                | 183                    | 2067                   | 8.13                                                |
| 6 | France                              | 70                                  | 79                                | 5                      | 63                     | 7.35                                                |
| 7 | France                              | 20                                  | 29                                | 44                     | 800                    | 5.21                                                |
| 8 | France                              | 10                                  | 19                                | 1                      | 21                     | 4.55                                                |
| 9 | France                              | 90                                  | 99                                | 0                      | 2                      | 0.00                                                |

#### Balance

|    | geography<br>character varying (20) | balance_range_start<br>numeric | balance_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|----|-------------------------------------|--------------------------------|------------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1  | France                              | 230000                         | 239999                       | 1                      | 0                      | 100.00                                              |
| 2  | France                              | 20000                          | 29999                        | 3                      | 0                      | 100.00                                              |
| 3  | France                              | 200000                         | 209999                       | 7                      | 3                      | 70.00                                               |
| 4  | France                              | 40000                          | 49999                        | 13                     | 16                     | 44.83                                               |
| 5  | France                              | 30000                          | 39999                        | 3                      | 4                      | 42.86                                               |
| 6  | France                              | 210000                         | 219999                       | 2                      | 3                      | 40.00                                               |
| 7  | France                              | 190000                         | 199999                       | 8                      | 15                     | 34.78                                               |
| 8  | France                              | 180000                         | 189999                       | 13                     | 26                     | 33.33                                               |
| 9  | France                              | 170000                         | 179999                       | 22                     | 48                     | 31.43                                               |
| 10 | France                              | 50000                          | 59999                        | 10                     | 27                     | 27.03                                               |
| 11 | France                              | 60000                          | 69999                        | 16                     | 56                     | 22.22                                               |
| 12 | France                              | 150000                         | 159999                       | 30                     | 130                    | 18.75                                               |
| 13 | France                              | 130000                         | 139999                       | 51                     | 233                    | 17.96                                               |
| 14 | France                              | 160000                         | 169999                       | 22                     | 102                    | 17.74                                               |
| 15 | France                              | 140000                         | 149999                       | 43                     | 203                    | 17.48                                               |
| 16 | France                              | 80000                          | 89999                        | 31                     | 151                    | 17.03                                               |
| 17 | France                              | 70000                          | 79999                        | 20                     | 99                     | 16.81                                               |
| 18 | France                              | 120000                         | 129999                       | 56                     | 285                    | 16.42                                               |
| 19 | France                              | 100000                         | 109999                       | 43                     | 244                    | 14.96                                               |
| 20 | France                              | 110000                         | 119999                       | 45                     | 273                    | 14.15                                               |
| 21 | France                              | 0                              | 9999                         | 338                    | 2081                   | 13.97                                               |
| 22 | France                              | 90000                          | 99999                        | 33                     | 204                    | 13.92                                               |
| 23 | France                              | 10000                          | 19999                        | 0                      | 1                      | 0.00                                                |

#### Credit Score

|   | geography<br>character varying (20) | creditscore_range<br>text | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|---------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | France                              | 300 - 399                 | 6                      | 0                      | 100.00                                              |
| 2 | France                              | 400 - 499                 | 58                     | 263                    | 18.07                                               |
| 3 | France                              | 500 - 599                 | 202                    | 1002                   | 16.78                                               |
| 4 | France                              | 600 - 699                 | 314                    | 1601                   | 16.40                                               |
| 5 | France                              | 800 - 899                 | 50                     | 273                    | 15.48                                               |
| 6 | France                              | 700 - 799                 | 180                    | 1065                   | 14.46                                               |

#### Gender

|   | geography<br>character varying (20) | gender<br>character varying (10) | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|----------------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | France                              | Female                           | 460                    | 1801                   | 20.34                                               |
| 2 | France                              | Male                             | 350                    | 2403                   | 12.71                                               |

#### Numofproducts

|   | geography<br>character varying (20) | numofproducts<br>integer | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|--------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | France                              | 4                        | 29                     | 0                      | 100.00                                              |
| 2 | France                              | 3                        | 82                     | 22                     | 78.85                                               |
| 3 | France                              | 1                        | 564                    | 1950                   | 22.43                                               |
| 4 | France                              | 2                        | 135                    | 2232                   | 5.70                                                |

for Balance:

```
SELECT geography, FLOOR(balance / 10000) * 10000 AS
balance_range_start, FLOOR(balance / 10000) * 10000 + 9999 AS
balance_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS
exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS
stayed_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) *
100.0 / COUNT(*)), 2) AS "exited percentage ((exit/exit+stay)*100)"
FROM bank_churn WHERE balance <= 300000 AND geography =
'France' GROUP BY geography, floor(balance/10000) ORDER BY
"exited percentage ((exit/exit+stay)*100)" DESC;
```

for Age:

```
SELECT geography, FLOOR(age / 10) * 10 AS age_range_start,
FLOOR(age / 10) * 10 + 9 AS age_range_end, COUNT(*) FILTER
(WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER
(WHERE exited = FALSE) AS stayed_count, ROUND((COUNT(*)
FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS
"exited percentage ((exit/exit+stay)*100)" FROM bank_churn
WHERE age <= 100 AND geography = 'France' GROUP BY
geography, floor(age/10) ORDER BY "exited percentage
((exit/exit+stay)*100)" DESC;
```

#### Estimated Salary

|    | geography<br>character varying (20) | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|----|-------------------------------------|-------------------------------|-----------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1  | France                              | 170000                        | 179999                      | 54                     | 208                    | 20.61                                               |
| 2  | France                              | 190000                        | 199999                      | 47                     | 205                    | 18.65                                               |
| 3  | France                              | 20000                         | 29999                       | 44                     | 206                    | 17.60                                               |
| 4  | France                              | 50000                         | 59999                       | 43                     | 204                    | 17.41                                               |
| 5  | France                              | 90000                         | 99999                       | 43                     | 204                    | 17.41                                               |
| 6  | France                              | 60000                         | 69999                       | 48                     | 228                    | 17.39                                               |
| 7  | France                              | 110000                        | 119999                      | 41                     | 195                    | 17.37                                               |
| 8  | France                              | 130000                        | 139999                      | 41                     | 200                    | 17.01                                               |
| 9  | France                              | 150000                        | 159999                      | 39                     | 191                    | 16.96                                               |
| 10 | France                              | 180000                        | 189999                      | 39                     | 196                    | 16.60                                               |
| 11 | France                              | 100000                        | 109999                      | 44                     | 223                    | 16.48                                               |
| 12 | France                              | 10000                         | 19999                       | 37                     | 190                    | 16.30                                               |
| 13 | France                              | 160000                        | 169999                      | 40                     | 208                    | 16.13                                               |
| 14 | France                              | 70000                         | 79999                       | 43                     | 224                    | 16.10                                               |
| 15 | France                              | 140000                        | 149999                      | 42                     | 224                    | 15.79                                               |
| 16 | France                              | 30000                         | 39999                       | 38                     | 211                    | 15.28                                               |
| 17 | France                              | 120000                        | 129999                      | 37                     | 221                    | 14.54                                               |
| 18 | France                              | 80000                         | 89999                       | 32                     | 228                    | 12.31                                               |
| 19 | France                              | 0                             | 9999                        | 31                     | 228                    | 12.06                                               |
| 20 | France                              | 40000                         | 49999                       | 27                     | 212                    | 11.30                                               |

for Credit Score:

```
SELECT geography, CONCAT(FLOOR(creditscore / 100) * 100,
' - ', FLOOR(creditscore / 100) * 100 + 99) AS creditscore_range,
COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
COUNT(*) FILTER (WHERE exited = FALSE) AS stayed_count,
ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 /
COUNT(*)), 2) AS "exited percentage ((exit/exit+stay)*100)"
FROM bank_churn WHERE creditscore <= 1000 AND geography =
'France' GROUP BY geography, floor(creditscore/100) ORDER BY
"exited percentage ((exit/exit+stay)*100)" DESC;
```

for Gender:

```
SELECT geography, gender, COUNT(*) FILTER (WHERE exited =
TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE)
AS stayed_count, ROUND((COUNT(*) FILTER (WHERE exited =
TRUE) * 100.0 / COUNT(*)), 2) AS "exited percentage
((exit/exit+stay)*100)" FROM bank_churn WHERE geography =
'France' GROUP BY geography, gender ORDER BY "exited percentage
((exit/exit+stay)*100)" DESC;
```

for numofproducts:

```
SELECT geography, numofproducts, COUNT(*) FILTER (WHERE
exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited =
FALSE) AS stayed_count, ROUND((COUNT(*) FILTER (WHERE exited =
TRUE) * 100.0 / COUNT(*)), 2) AS "exited percentage
((exit/exit+stay)*100)" FROM bank_churn WHERE geography =
'France' GROUP BY geography, numofproducts ORDER BY "exited
percentage ((exit/exit+stay)*100)" DESC;
```

for Estimated Salary:

```
SELECT geography, FLOOR(estimatedsalary / 10000) * 10000 AS
salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999
AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS
exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS
stayed_count, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) *
100.0 / COUNT(*)), 2) AS "exited percentage ((exit/exit+stay)*100)"
FROM bank_churn WHERE estimatedsalary <= 300000 AND
geography = 'France' GROUP BY geography, FLOOR(estimatedsalary
/ 10000) ORDER BY "exited percentage ((exit/exit+stay)*100)" DESC;
```

## Germany

## Age

|   | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|-------------------------------------|-----------------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | Germany                             | 50                                  | 59                                | 194                    | 83                     | 70.04                                               |
| 2 | Germany                             | 60                                  | 69                                | 45                     | 39                     | 53.57                                               |
| 3 | Germany                             | 40                                  | 49                                | 334                    | 406                    | 45.14                                               |
| 4 | Germany                             | 30                                  | 39                                | 188                    | 609                    | 38.66                                               |
| 5 | Germany                             | 70                                  | 79                                | 6                      | 30                     | 16.67                                               |
| 6 | Germany                             | 20                                  | 29                                | 46                     | 312                    | 12.85                                               |
| 7 | Germany                             | 10                                  | 19                                | 1                      | 13                     | 7.14                                                |
| 8 | Germany                             | 80                                  | 89                                | 0                      | 3                      | 0.00                                                |

## Balance

|    | geography<br>character varying (20) | balance_range_start<br>numeric | balance_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|----|-------------------------------------|--------------------------------|------------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1  | Germany                             | 110000                         | 119999                       | 172                    | 199                    | 46.36                                               |
| 2  | Germany                             | 100000                         | 109999                       | 144                    | 208                    | 40.91                                               |
| 3  | Germany                             | 120000                         | 129999                       | 146                    | 237                    | 38.12                                               |
| 4  | Germany                             | 130000                         | 139999                       | 109                    | 205                    | 34.71                                               |
| 5  | Germany                             | 140000                         | 149999                       | 75                     | 157                    | 32.33                                               |
| 6  | Germany                             | 90000                          | 99999                        | 68                     | 173                    | 28.22                                               |
| 7  | Germany                             | 150000                         | 159999                       | 38                     | 107                    | 26.21                                               |
| 8  | Germany                             | 160000                         | 169999                       | 19                     | 64                     | 22.89                                               |
| 9  | Germany                             | 80000                          | 89999                        | 23                     | 97                     | 19.17                                               |
| 10 | Germany                             | 70000                          | 79999                        | 13                     | 80                     | 13.98                                               |
| 11 | Germany                             | 60000                          | 69999                        | 4                      | 40                     | 9.09                                                |
| 12 | Germany                             | 170000                         | 179999                       | 2                      | 39                     | 4.88                                                |
| 13 | Germany                             | 180000                         | 189999                       | 1                      | 30                     | 3.23                                                |
| 14 | Germany                             | 210000                         | 219999                       | 0                      | 2                      | 0.00                                                |
| 15 | Germany                             | 30000                          | 39999                        | 0                      | 8                      | 0.00                                                |
| 16 | Germany                             | 40000                          | 49999                        | 0                      | 9                      | 0.00                                                |
| 17 | Germany                             | 200000                         | 209999                       | 0                      | 4                      | 0.00                                                |
| 18 | Germany                             | 50000                          | 59999                        | 0                      | 26                     | 0.00                                                |
| 19 | Germany                             | 20000                          | 29999                        | 0                      | 2                      | 0.00                                                |
| 20 | Germany                             | 190000                         | 199999                       | 0                      | 8                      | 0.00                                                |

## Gender

|   | geography<br>character varying (20) | gender<br>character varying (10) | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|----------------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | Germany                             | Female                           | 448                    | 745                    | 37.55                                               |
| 2 | Germany                             | Male                             | 366                    | 950                    | 27.81                                               |

## Spain

## Age

|   | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|-------------------------------------|-----------------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | Spain                               | 50                                  | 59                                | 96                     | 114                    | 45.71                                               |
| 2 | Spain                               | 60                                  | 69                                | 25                     | 70                     | 26.32                                               |
| 3 | Spain                               | 40                                  | 49                                | 155                    | 481                    | 24.37                                               |
| 4 | Spain                               | 70                                  | 79                                | 3                      | 29                     | 9.38                                                |
| 5 | Spain                               | 30                                  | 39                                | 102                    | 997                    | 9.28                                                |
| 6 | Spain                               | 20                                  | 29                                | 31                     | 358                    | 7.95                                                |
| 7 | Spain                               | 10                                  | 19                                | 1                      | 12                     | 7.69                                                |
| 8 | Spain                               | 80                                  | 89                                | 0                      | 2                      | 0.00                                                |

## Balance

|    | geography<br>character varying (20) | balance_range_start<br>numeric | balance_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|----|-------------------------------------|--------------------------------|------------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1  | Spain                               | 250000                         | 259999                       | 1                      | 0                      | 100.00                                              |
| 2  | Spain                               | 210000                         | 219999                       | 2                      | 0                      | 100.00                                              |
| 3  | Spain                               | 200000                         | 209999                       | 5                      | 2                      | 71.43                                               |
| 4  | Spain                               | 20000                          | 29999                        | 2                      | 1                      | 66.67                                               |
| 5  | Spain                               | 220000                         | 229999                       | 1                      | 1                      | 50.00                                               |
| 6  | Spain                               | 10000                          | 19999                        | 1                      | 1                      | 50.00                                               |
| 7  | Spain                               | 180000                         | 189999                       | 8                      | 8                      | 50.00                                               |
| 8  | Spain                               | 30000                          | 39999                        | 1                      | 1                      | 50.00                                               |
| 9  | Spain                               | 30000                          | 59999                        | 8                      | 9                      | 47.06                                               |
| 10 | Spain                               | 60000                          | 69999                        | 17                     | 23                     | 42.50                                               |
| 11 | Spain                               | 170000                         | 179999                       | 13                     | 35                     | 27.08                                               |
| 12 | Spain                               | 40000                          | 49999                        | 2                      | 6                      | 25.00                                               |
| 13 | Spain                               | 70000                          | 79999                        | 15                     | 47                     | 24.19                                               |
| 14 | Spain                               | 190000                         | 199999                       | 2                      | 7                      | 22.22                                               |
| 15 | Spain                               | 150000                         | 159999                       | 17                     | 64                     | 20.99                                               |
| 16 | Spain                               | 80000                          | 89999                        | 20                     | 78                     | 20.41                                               |
| 17 | Spain                               | 90000                          | 99999                        | 22                     | 99                     | 18.18                                               |
| 18 | Spain                               | 160000                         | 169999                       | 10                     | 47                     | 17.54                                               |
| 19 | Spain                               | 130000                         | 139999                       | 23                     | 113                    | 16.91                                               |
| 20 | Spain                               | 110000                         | 119999                       | 23                     | 120                    | 16.08                                               |
| 21 | Spain                               | 140000                         | 149999                       | 15                     | 87                     | 14.71                                               |
| 22 | Spain                               | 0                              | 9999                         | 163                    | 1036                   | 13.59                                               |
| 23 | Spain                               | 120000                         | 129999                       | 23                     | 151                    | 13.22                                               |
| 24 | Spain                               | 100000                         | 109999                       | 19                     | 128                    | 12.93                                               |

## Gender

|   | geography<br>character varying (20) | gender<br>character varying (10) | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|----------------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | Spain                               | Female                           | 231                    | 858                    | 21.21                                               |
| 2 | Spain                               | Male                             | 182                    | 1206                   | 13.11                                               |

## Estimated Salary

|    | geography<br>character varying (20) | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|----|-------------------------------------|-------------------------------|-----------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1  | Germany                             | 40000                         | 49999                       | 52                     | 79                     | 39.69                                               |
| 2  | Germany                             | 0                             | 9999                        | 50                     | 77                     | 39.37                                               |
| 3  | Germany                             | 60000                         | 69999                       | 46                     | 73                     | 38.66                                               |
| 4  | Germany                             | 10000                         | 19999                       | 40                     | 70                     | 36.36                                               |
| 5  | Germany                             | 80000                         | 89999                       | 41                     | 72                     | 36.28                                               |
| 6  | Germany                             | 110000                        | 119999                      | 41                     | 75                     | 35.34                                               |
| 7  | Germany                             | 180000                        | 189999                      | 45                     | 83                     | 35.16                                               |
| 8  | Germany                             | 140000                        | 149999                      | 43                     | 86                     | 33.83                                               |
| 9  | Germany                             | 160000                        | 169999                      | 41                     | 83                     | 33.06                                               |
| 10 | Germany                             | 20000                         | 29999                       | 42                     | 87                     | 32.56                                               |
| 11 | Germany                             | 120000                        | 129999                      | 40                     | 85                     | 32.00                                               |
| 12 | Germany                             | 30000                         | 39999                       | 36                     | 83                     | 30.25                                               |
| 13 | Germany                             | 90000                         | 99999                       | 35                     | 81                     | 30.17                                               |
| 14 | Germany                             | 130000                        | 139999                      | 38                     | 90                     | 29.69                                               |
| 15 | Germany                             | 190000                        | 199999                      | 41                     | 100                    | 29.08                                               |
| 16 | Germany                             | 70000                         | 79999                       | 38                     | 93                     | 29.01                                               |
| 17 | Germany                             | 170000                        | 179999                      | 42                     | 106                    | 28.38                                               |
| 18 | Germany                             | 100000                        | 109999                      | 40                     | 102                    | 28.17                                               |
| 19 | Germany                             | 150000                        | 159999                      | 27                     | 68                     | 28.13                                               |
| 20 | Germany                             | 50000                         | 59999                       | 36                     | 101                    | 26.28                                               |

## Credit Score

|   | geography<br>character varying (20) | creditscore_range<br>text | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|---------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | Germany                             | 300 - 399                 | 6                      | 0                      | 100.00                                              |
| 2 | Germany                             | 500 - 599                 | 203                    | 385                    | 34.52                                               |
| 3 | Germany                             | 700 - 799                 | 213                    | 420                    | 33.65                                               |
| 4 | Germany                             | 400 - 499                 | 55                     | 111                    | 33.13                                               |
| 5 | Germany                             | 600 - 699                 | 285                    | 655                    | 30.32                                               |
| 6 | Germany                             | 800 - 899                 | 52                     | 124                    | 29.55                                               |

## Numofproducts

|   | geography<br>character varying (20) | numofproducts<br>integer | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|--------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | Germany                             |                          | 4                      | 24                     | 100.00                                              |
| 2 | Germany                             |                          | 3                      | 86                     | 89.58                                               |
| 3 | Germany                             |                          | 1                      | 578                    | 42.85                                               |
| 4 | Germany                             |                          | 2                      | 126                    | 91.4                                                |

## Estimated Salary

|    | geography<br>character varying (20) | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint | Exit Percentage ((Exit/Exit+Stay)*100)<br>numeric |
|----|-------------------------------------|-------------------------------|-----------------------------|------------------------|------------------------|---------------------------------------------------|
| 1  | Spain                               | 170000                        | 179999                      | 27                     | 94                     | 22.31                                             |
| 2  | Spain                               | 150000                        | 159999                      | 30                     | 105                    | 22.22                                             |
| 3  | Spain                               | 140000                        | 149999                      | 25                     | 101                    | 19.84                                             |
| 4  | Spain                               | 160000                        | 169999                      | 21                     | 85                     | 19.81                                             |
| 5  | Spain                               | 80000                         | 89999                       | 26                     | 111                    | 18.96                                             |
| 6  | Spain                               | 60000                         | 69999                       | 20                     | 92                     | 17.86                                             |
| 7  | Spain                               | 130000                        | 139999                      | 23                     | 110                    | 17.29                                             |
| 8  | Spain                               | 110000                        | 119999                      | 24                     | 115                    | 17.27                                             |
| 9  | Spain                               | 100000                        | 109999                      | 21                     | 106                    | 16.54                                             |
| 10 | Spain                               | 30000                         | 39999                       | 18                     | 91                     | 16.51                                             |
| 11 | Spain                               | 0                             | 9999                        | 20                     | 104                    | 16.13                                             |
| 12 | Spain                               | 20000                         | 29999                       | 18                     | 95                     | 15.93                                             |
| 13 | Spain                               | 180000                        | 189999                      | 17                     | 96                     | 15.04                                             |
| 14 | Spain                               | 90000                         | 99999                       | 19                     | 110                    | 14.73                                             |
| 15 | Spain                               | 190000                        | 199999                      | 17                     | 99                     | 14.46                                             |
| 16 | Spain                               | 10000                         | 19999                       | 20                     | 121                    | 14.18                                             |
| 17 | Spain                               | 70000                         | 79999                       | 17                     | 105                    | 13.93                                             |
| 18 | Spain                               | 120000                        | 129999                      | 17                     | 105                    | 13.93                                             |
| 19 | Spain                               | 50000                         | 59999                       | 17                     | 107                    | 13.71                                             |
| 20 | Spain                               | 40000                         | 49999                       | 16                     | 112                    | 12.50                                             |

## Credit Score

|   | geography<br>character varying (20) | creditscore_range<br>text | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|---------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | Spain                               | 300 - 399                 | 7                      | 0                      | 100.00                                              |
| 2 | Spain                               | 500 - 599                 | 105                    | 505                    | 17.21                                               |
| 3 | Spain                               | 700 - 799                 | 103                    | 512                    | 16.75                                               |
| 4 | Spain                               | 800 - 899                 | 26                     | 130                    | 16.67                                               |
| 5 | Spain                               | 600 - 699                 | 154                    | 809                    | 15.99                                               |
| 6 | Spain                               | 400 - 499                 | 18                     | 108                    | 14.29                                               |

## Numofproducts

|   | geography<br>character varying (20) | numofproducts<br>integer | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100)<br>numeric |
|---|-------------------------------------|--------------------------|------------------------|------------------------|-----------------------------------------------------|
| 1 | Spain                               |                          | 4                      | 7                      | 100.00                                              |
| 2 | Spain                               |                          | 3                      | 52                     | 78.79                                               |
| 3 | Spain                               |                          | 1                      | 267                    | 21.87                                               |
| 4 | Spain                               |                          | 2                      | 87                     | 7.35                                                |

## SQL Analysis & Key Findings: Data Exploration - LEVEL 2 - Combined Attribute Analysis

### Geography + Age + Balance + Estimated Salary

```
SELECT geography, FLOOR(age / 10) * 10 AS age_range_start, FLOOR(age / 10) * 10 + 9 AS age_range_end, FLOOR(balance / 10000) * 10000 AS balance_range_start, FLOOR(balance / 10000) * 10000 + 9999 AS balance_range_end, FLOOR(estimatedsalary / 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count(*) as total_customers, ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage FROM bank_churn WHERE age <= 100 and balance <=300000 and estimatedsalary <=300000 GROUP BY FLOOR(estimatedsalary / 10000), floor(age/10), floor(balance/10000), geography ORDER BY churn_percentage desc;
```

|    | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | balance_range_start<br>numeric | balance_range_end<br>numeric | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|-------------------------------------|-------------------------------------|-----------------------------------|--------------------------------|------------------------------|-------------------------------|-----------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | Spain                               | 50                                  | 59                                | 80000                          | 89999                        | 80000                         | 89999                       | 1                      | 0                        | 1                         | 100.00                      |
| 2  | France                              | 50                                  | 59                                | 100000                         | 109999                       | 10000                         | 19999                       | 1                      | 0                        | 1                         | 100.00                      |
| 3  | France                              | 50                                  | 59                                | 120000                         | 129999                       | 170000                        | 179999                      | 2                      | 0                        | 2                         | 100.00                      |
| 4  | Germany                             | 50                                  | 59                                | 150000                         | 159999                       | 80000                         | 89999                       | 1                      | 0                        | 1                         | 100.00                      |
| 5  | Spain                               | 60                                  | 69                                | 0                              | 9999                         | 30000                         | 39999                       | 1                      | 0                        | 1                         | 100.00                      |
| 6  | France                              | 20                                  | 29                                | 180000                         | 189999                       | 110000                        | 119999                      | 1                      | 0                        | 1                         | 100.00                      |
| 7  | Germany                             | 20                                  | 29                                | 130000                         | 139999                       | 180000                        | 189999                      | 1                      | 0                        | 1                         | 100.00                      |
| 8  | Spain                               | 40                                  | 49                                | 60000                          | 69999                        | 130000                        | 139999                      | 1                      | 0                        | 1                         | 100.00                      |
| 9  | Germany                             | 60                                  | 69                                | 140000                         | 149999                       | 190000                        | 199999                      | 2                      | 0                        | 2                         | 100.00                      |
| 10 | Germany                             | 50                                  | 59                                | 110000                         | 119999                       | 160000                        | 169999                      | 1                      | 0                        | 1                         | 100.00                      |
| 11 | Germany                             | 60                                  | 69                                | 100000                         | 109999                       | 180000                        | 189999                      | 1                      | 0                        | 1                         | 100.00                      |
| 12 | Spain                               | 50                                  | 59                                | 200000                         | 209999                       | 150000                        | 159999                      | 1                      | 0                        | 1                         | 100.00                      |
| 13 | France                              | 60                                  | 69                                | 70000                          | 79999                        | 160000                        | 169999                      | 1                      | 0                        | 1                         | 100.00                      |
| 14 | Spain                               | 50                                  | 59                                | 0                              | 9999                         | 180000                        | 189999                      | 2                      | 0                        | 2                         | 100.00                      |
| 15 | Germany                             | 30                                  | 39                                | 150000                         | 159999                       | 190000                        | 199999                      | 2                      | 0                        | 2                         | 100.00                      |
| 16 | Germany                             | 50                                  | 59                                | 160000                         | 169999                       | 20000                         | 29999                       | 1                      | 0                        | 1                         | 100.00                      |
| 17 | Germany                             | 40                                  | 49                                | 130000                         | 139999                       | 120000                        | 129999                      | 2                      | 0                        | 2                         | 100.00                      |
| 18 | Spain                               | 50                                  | 59                                | 130000                         | 139999                       | 60000                         | 69999                       | 1                      | 0                        | 1                         | 100.00                      |
| 19 | Germany                             | 50                                  | 59                                | 90000                          | 99999                        | 40000                         | 49999                       | 1                      | 0                        | 1                         | 100.00                      |
| 20 | France                              | 50                                  | 59                                | 150000                         | 159999                       | 150000                        | 159999                      | 1                      | 0                        | 1                         | 100.00                      |
| 21 | Germany                             | 60                                  | 69                                | 120000                         | 129999                       | 90000                         | 99999                       | 1                      | 0                        | 1                         | 100.00                      |
| 22 | Germany                             | 20                                  | 29                                | 70000                          | 79999                        | 120000                        | 129999                      | 1                      | 0                        | 1                         | 100.00                      |
| 23 | Germany                             | 40                                  | 49                                | 150000                         | 159999                       | 50000                         | 59999                       | 2                      | 0                        | 2                         | 100.00                      |
| 24 | Germany                             | 50                                  | 59                                | 110000                         | 119999                       | 160000                        | 169999                      | 4                      | 0                        | 4                         | 100.00                      |
| 25 | Germany                             | 50                                  | 59                                | 140000                         | 149999                       | 120000                        | 129999                      | 1                      | 0                        | 1                         | 100.00                      |
| 26 | Spain                               | 20                                  | 29                                | 160000                         | 169999                       | 150000                        | 159999                      | 1                      | 0                        | 1                         | 100.00                      |
| 27 | France                              | 30                                  | 39                                | 180000                         | 189999                       | 60000                         | 69999                       | 2                      | 0                        | 2                         | 100.00                      |
| 28 | Spain                               | 50                                  | 59                                | 140000                         | 149999                       | 0                             | 9999                        | 1                      | 0                        | 1                         | 100.00                      |
| 29 | Spain                               | 50                                  | 59                                | 180000                         | 189999                       | 110000                        | 119999                      | 1                      | 0                        | 1                         | 100.00                      |
| 30 | Spain                               | 50                                  | 59                                | 130000                         | 139999                       | 180000                        | 189999                      | 1                      | 0                        | 1                         | 100.00                      |

| Showing rows: 1 to 2837 |                                     |                                     |                                   |                                |                              |                               |                             |                        |                          |                           |                             |
|-------------------------|-------------------------------------|-------------------------------------|-----------------------------------|--------------------------------|------------------------------|-------------------------------|-----------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| Page No: 1 of 14        |                                     |                                     |                                   |                                |                              |                               |                             |                        |                          |                           |                             |
|                         | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | balance_range_start<br>numeric | balance_range_end<br>numeric | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
| 431                     | France                              | 50                                  | 59                                | 0                              | 9999                         | 190000                        | 199999                      | 7                      | 1                        | 8                         | 87.50                       |
| 432                     | Germany                             | 50                                  | 59                                | 120000                         | 129999                       | 10000                         | 19999                       | 6                      | 1                        | 7                         | 85.71                       |
| 433                     | Germany                             | 40                                  | 49                                | 110000                         | 119999                       | 120000                        | 129999                      | 5                      | 1                        | 6                         | 83.33                       |
| 434                     | Germany                             | 40                                  | 49                                | 100000                         | 109999                       | 10000                         | 19999                       | 5                      | 1                        | 6                         | 83.33                       |
| 435                     | Germany                             | 50                                  | 59                                | 120000                         | 129999                       | 130000                        | 139999                      | 4                      | 1                        | 5                         | 80.00                       |
| 436                     | Germany                             | 40                                  | 49                                | 110000                         | 119999                       | 20000                         | 29999                       | 4                      | 1                        | 5                         | 80.00                       |
| 437                     | France                              | 50                                  | 59                                | 0                              | 9999                         | 150000                        | 159999                      | 4                      | 1                        | 5                         | 80.00                       |
| 438                     | Germany                             | 50                                  | 59                                | 120000                         | 129999                       | 80000                         | 89999                       | 4                      | 1                        | 5                         | 80.00                       |
| 439                     | Germany                             | 40                                  | 49                                | 140000                         | 149999                       | 180000                        | 189999                      | 4                      | 1                        | 5                         | 80.00                       |
| 440                     | Germany                             | 40                                  | 49                                | 130000                         | 139999                       | 20000                         | 29999                       | 4                      | 1                        | 5                         | 80.00                       |
| 441                     | Germany                             | 40                                  | 49                                | 140000                         | 149999                       | 30000                         | 39999                       | 3                      | 1                        | 4                         | 75.00                       |
| 442                     | Germany                             | 50                                  | 59                                | 100000                         | 109999                       | 20000                         | 29999                       | 3                      | 1                        | 4                         | 75.00                       |
| 443                     | Germany                             | 40                                  | 49                                | 110000                         | 119999                       | 40000                         | 49999                       | 3                      | 1                        | 4                         | 75.00                       |
| 444                     | France                              | 40                                  | 49                                | 120000                         | 129999                       | 30000                         | 39999                       | 3                      | 1                        | 4                         | 75.00                       |
| 445                     | Germany                             | 40                                  | 49                                | 80000                          | 89999                        | 80000                         | 89999                       | 3                      | 1                        | 4                         | 75.00                       |
| 446                     | Germany                             | 50                                  | 59                                | 100000                         | 109999                       | 50000                         | 59999                       | 3                      | 1                        | 4                         | 75.00                       |
| 447                     | Germany                             | 50                                  | 59                                | 120000                         | 129999                       | 0                             | 9999                        | 3                      | 1                        | 4                         | 75.00                       |
| 448                     | Germany                             | 50                                  | 59                                | 140000                         | 149999                       | 40000                         | 49999                       | 3                      | 1                        | 4                         | 75.00                       |
| 449                     | Germany                             | 40                                  | 49                                | 110000                         | 119999                       | 130000                        | 139999                      | 6                      | 2                        | 8                         | 75.00                       |
| 450                     | Germany                             | 40                                  | 49                                | 120000                         | 129999                       | 60000                         | 69999                       | 6                      | 2                        | 8                         | 75.00                       |
| 451                     | Germany                             | 50                                  | 59                                | 100000                         | 109999                       | 160000                        | 169999                      | 3                      | 1                        | 4                         | 75.00                       |
| 452                     | Spain                               | 50                                  | 59                                | 0                              | 9999                         | 70000                         | 79999                       | 3                      | 1                        | 4                         | 75.00                       |
| 453                     | France                              | 50                                  | 59                                | 0                              | 9999                         | 100000                        | 109999                      | 6                      | 2                        | 8                         | 75.00                       |
| 454                     | Germany                             | 40                                  | 49                                | 110000                         | 119999                       | 160000                        | 169999                      | 3                      | 1                        | 4                         | 75.00                       |
| 455                     | Germany                             | 40                                  | 49                                | 120000                         | 129999                       | 80000                         | 89999                       | 3                      | 1                        | 4                         | 75.00                       |
| 456                     | Germany                             | 30                                  | 39                                | 110000                         | 119999                       | 40000                         | 49999                       | 6                      | 2                        | 8                         | 75.00                       |
| 457                     | Germany                             | 20                                  | 29                                | 110000                         | 119999                       | 110000                        | 119999                      | 3                      | 1                        | 4                         | 75.00                       |
| 458                     | Germany                             | 50                                  | 59                                | 110000                         | 119999                       | 40000                         | 49999                       | 3                      | 1                        | 4                         | 75.00                       |
| 459                     | Germany                             | 40                                  | 49                                | 100000                         | 109999                       | 180000                        | 189999                      | 8                      | 3                        | 11                        | 72.73                       |
| 460                     | Germany                             | 40                                  | 49                                | 120000                         | 129999                       | 0                             | 9999                        | 5                      | 2                        | 7                         | 71.43                       |

| Showing rows: 1 to 2837 |                                     |                                     |                                   |                                |                              |                               |                             |                        |                          |                           |                             |
|-------------------------|-------------------------------------|-------------------------------------|-----------------------------------|--------------------------------|------------------------------|-------------------------------|-----------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| Page No: 1              |                                     |                                     |                                   |                                |                              |                               |                             |                        |                          |                           |                             |
|                         | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | balance_range_start<br>numeric | balance_range_end<br>numeric | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
| 1                       | France                              | 40                                  | 49                                | 0                              | 9999                         | 110000                        | 119999                      | 12                     | 17                       | 29                        |                             |
| 2                       | France                              | 40                                  | 49                                | 0                              | 9999                         | 100000                        | 109999                      | 11                     | 27                       | 38                        |                             |
| 3                       | France                              | 30                                  | 39                                | 0                              | 9999                         | 50000                         | 59999                       | 10                     | 46                       | 56                        |                             |
| 4                       | France                              | 50                                  | 59                                | 0                              | 9999                         | 170000                        | 179999                      | 9                      | 5                        | 14                        |                             |
| 5                       | France                              | 40                                  | 49                                | 0                              | 9999                         | 70000                         | 79999                       | 9                      | 25                       | 34                        |                             |
| 6                       | France                              | 40                                  | 49                                | 0                              | 9999                         | 140000                        | 149999                      | 9                      | 32                       | 41                        |                             |
| 7                       | Germany                             | 40                                  | 49                                | 100000                         | 109999                       | 180000                        | 189999                      | 8                      | 3                        | 11                        |                             |
| 8                       | France                              | 40                                  | 49                                | 0                              | 9999                         | 170000                        | 179999                      | 8                      | 31                       | 39                        |                             |
| 9                       | France                              | 40                                  | 49                                | 0                              | 9999                         | 130000                        | 139999                      | 8                      | 25                       | 33                        |                             |
| 10                      | France                              | 40                                  | 49                                | 0                              | 9999                         | 80000                         | 89999                       | 8                      | 23                       | 31                        |                             |
| 11                      | France                              | 40                                  | 49                                | 0                              | 9999                         | 190000                        | 199999                      | 8                      | 21                       | 29                        |                             |
| 12                      | France                              | 40                                  | 49                                | 0                              | 9999                         | 60000                         | 69999                       | 8                      | 29                       | 37                        |                             |
| 13                      | France                              | 30                                  | 39                                | 0                              | 9999                         | 170000                        | 179999                      | 7                      | 37                       | 44                        |                             |
| 14                      | France                              | 50                                  | 59                                | 0                              | 9999                         | 190000                        | 199999                      | 7                      | 1                        | 8                         |                             |
| 15                      | France                              | 40                                  | 49                                | 0                              | 9999                         | 90000                         | 99999                       | 7                      | 25                       | 32                        |                             |
| 16                      | Germany                             | 40                                  | 49                                | 120000                         | 129999                       | 60000                         | 69999                       | 6                      | 2                        | 8                         |                             |
| 17                      | France                              | 50                                  | 59                                | 0                              | 9999                         | 90000                         | 99999                       | 6                      | 6                        | 12                        |                             |
| 18                      | France                              | 40                                  | 49                                | 0                              | 9999                         | 10000                         | 19999                       | 6                      | 17                       | 23                        |                             |
| 19                      | France                              | 30                                  | 39                                | 0                              | 9999                         | 180000                        | 189999                      | 6                      | 48                       | 54                        |                             |

## Geography + Age + Estimated Salary

```
SELECT geography, FLOOR(age / 10) * 10 AS age_range_start, FLOOR(age / 10) * 10 + 9 AS age_range_end,
 FLOOR(estimatedsalary / 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999
 AS salary_range_end, COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count,
 COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count(*) as total_customers,
 ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage
FROM bank_churn WHERE age <= 100 and estimatedsalary <= 300000 GROUP BY FLOOR(estimatedsalary / 10000), floor(age/10), geography ORDER BY churn_percentage
desc;
```

|    | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|-------------------------------------|-------------------------------------|-----------------------------------|-------------------------------|-----------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | France                              | 40                                  | 49                                | 100000                        | 109999                      | 23                     |                          | 59                        |                             |
| 2  | France                              | 40                                  | 49                                | 110000                        | 119999                      | 21                     |                          | 38                        |                             |
| 3  | France                              | 30                                  | 39                                | 50000                         | 59999                       | 21                     |                          | 85                        |                             |
| 4  | Germany                             | 40                                  | 49                                | 180000                        | 189999                      | 21                     |                          | 17                        |                             |
| 5  | France                              | 40                                  | 49                                | 160000                        | 169999                      | 21                     |                          | 47                        |                             |
| 6  | France                              | 40                                  | 49                                | 60000                         | 69999                       | 20                     |                          | 56                        |                             |
| 7  | France                              | 40                                  | 49                                | 130000                        | 139999                      | 20                     |                          | 53                        |                             |
| 8  | France                              | 40                                  | 49                                | 140000                        | 149999                      | 19                     |                          | 57                        |                             |
| 9  | Germany                             | 40                                  | 49                                | 20000                         | 29999                       | 19                     |                          | 11                        |                             |
| 10 | Germany                             | 40                                  | 49                                | 0                             | 9999                        | 19                     |                          | 19                        |                             |
| 11 | Germany                             | 40                                  | 49                                | 30000                         | 39999                       | 19                     |                          | 24                        |                             |
| 12 | Germany                             | 40                                  | 49                                | 120000                        | 129999                      | 19                     |                          | 19                        |                             |
| 13 | France                              | 40                                  | 49                                | 170000                        | 179999                      | 19                     |                          | 56                        |                             |
| 14 | Germany                             | 40                                  | 49                                | 40000                         | 49999                       | 19                     |                          | 18                        |                             |
| 15 | Germany                             | 40                                  | 49                                | 130000                        | 139999                      | 18                     |                          | 17                        |                             |
| 16 | France                              | 40                                  | 49                                | 30000                         | 39999                       | 18                     |                          | 43                        |                             |
| 17 | Germany                             | 40                                  | 49                                | 140000                        | 149999                      | 18                     |                          | 15                        |                             |
| 18 | France                              | 40                                  | 49                                | 90000                         | 99999                       | 17                     |                          | 47                        |                             |
| 19 | Germany                             | 40                                  | 49                                | 100000                        | 109999                      | 17                     |                          | 27                        |                             |
| 20 | Germany                             | 40                                  | 49                                | 60000                         | 69999                       | 17                     |                          | 23                        |                             |
| 21 | France                              | 40                                  | 49                                | 190000                        | 199999                      | 17                     |                          | 43                        |                             |
| 22 | Germany                             | 40                                  | 49                                | 160000                        | 169999                      | 17                     |                          | 25                        |                             |
| 23 | France                              | 40                                  | 49                                | 70000                         | 79999                       | 16                     |                          | 62                        |                             |
| 24 | Germany                             | 40                                  | 49                                | 190000                        | 199999                      | 16                     |                          | 30                        |                             |
| 25 | Germany                             | 40                                  | 49                                | 80000                         | 89999                       | 16                     |                          | 16                        |                             |
| 26 | Germany                             | 30                                  | 39                                | 110000                        | 119999                      | 15                     |                          | 31                        |                             |
| 27 | Spain                               | 40                                  | 49                                | 150000                        | 159999                      | 15                     |                          | 23                        |                             |

Showing rows: 1 to 399

|    | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|-------------------------------------|-------------------------------------|-----------------------------------|-------------------------------|-----------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | Spain                               | 60                                  | 69                                | 140000                        | 149999                      | 1                      | 0                        | 1                         | 100.00                      |
| 2  | Spain                               | 60                                  | 69                                | 120000                        | 129999                      | 3                      | 0                        | 3                         | 100.00                      |
| 3  | Germany                             | 60                                  | 69                                | 150000                        | 159999                      | 4                      | 0                        | 4                         | 100.00                      |
| 4  | Germany                             | 10                                  | 19                                | 140000                        | 149999                      | 1                      | 0                        | 1                         | 100.00                      |
| 5  | France                              | 50                                  | 59                                | 190000                        | 199999                      | 14                     | 1                        | 15                        | 93.33                       |
| 6  | Germany                             | 50                                  | 59                                | 140000                        | 149999                      | 11                     | 1                        | 12                        | 91.67                       |
| 7  | France                              | 50                                  | 59                                | 150000                        | 159999                      | 14                     | 2                        | 16                        | 87.50                       |
| 8  | Germany                             | 50                                  | 59                                | 160000                        | 169999                      | 12                     | 2                        | 14                        | 85.71                       |
| 9  | Spain                               | 50                                  | 59                                | 140000                        | 149999                      | 11                     | 2                        | 13                        | 84.62                       |
| 10 | Germany                             | 50                                  | 59                                | 40000                         | 49999                       | 14                     | 3                        | 17                        | 82.35                       |
| 11 | Germany                             | 50                                  | 59                                | 0                             | 9999                        | 14                     | 3                        | 17                        | 82.35                       |
| 12 | Germany                             | 50                                  | 59                                | 120000                        | 129999                      | 9                      | 2                        | 11                        | 81.82                       |
| 13 | Germany                             | 60                                  | 69                                | 10000                         | 19999                       | 4                      | 1                        | 5                         | 80.00                       |
| 14 | Germany                             | 60                                  | 69                                | 80000                         | 89999                       | 4                      | 1                        | 5                         | 80.00                       |
| 15 | Germany                             | 50                                  | 59                                | 60000                         | 69999                       | 8                      | 2                        | 10                        | 80.00                       |
| 16 | Germany                             | 50                                  | 59                                | 90000                         | 99999                       | 11                     | 3                        | 14                        | 78.57                       |
| 17 | Spain                               | 50                                  | 59                                | 180000                        | 189999                      | 7                      | 2                        | 9                         | 77.78                       |
| 18 | Germany                             | 50                                  | 59                                | 150000                        | 159999                      | 6                      | 2                        | 8                         | 75.00                       |
| 19 | Germany                             | 60                                  | 69                                | 60000                         | 69999                       | 3                      | 1                        | 4                         | 75.00                       |
| 20 | Germany                             | 60                                  | 69                                | 190000                        | 199999                      | 3                      | 1                        | 4                         | 75.00                       |
| 21 | Germany                             | 50                                  | 59                                | 10000                         | 19999                       | 11                     | 4                        | 15                        | 73.33                       |
| 22 | Spain                               | 60                                  | 69                                | 170000                        | 179999                      | 5                      | 2                        | 7                         | 71.43                       |
| 23 | Spain                               | 50                                  | 59                                | 90000                         | 99999                       | 5                      | 2                        | 7                         | 71.43                       |
| 24 | Germany                             | 50                                  | 59                                | 70000                         | 79999                       | 12                     | 5                        | 17                        | 70.59                       |
| 25 | Germany                             | 50                                  | 59                                | 130000                        | 139999                      | 12                     | 5                        | 17                        | 70.59                       |
| 26 | France                              | 60                                  | 69                                | 30000                         | 39999                       | 7                      | 3                        | 10                        | 70.00                       |
| 27 | Germany                             | 50                                  | 59                                | 170000                        | 179999                      | 10                     | 5                        | 15                        | 66.67                       |
| 28 | Spain                               | 50                                  | 59                                | 160000                        | 169999                      | 4                      | 2                        | 6                         | 66.67                       |
| 29 | Germany                             | 50                                  | 59                                | 180000                        | 189999                      | 8                      | 4                        | 12                        | 66.67                       |
| 30 | Germany                             | 60                                  | 69                                | 140000                        | 149999                      | 4                      | 2                        | 6                         | 66.67                       |

## Geography + Estimated Salary

```
SELECT geography, FLOOR(estimatedsalary / 10000) * 10000 AS salary_range_start, FLOOR(estimatedsalary / 10000) * 10000 + 9999 AS salary_range_end,
 COUNT(*) FILTER (WHERE exited = TRUE) AS exited_count, COUNT(*) FILTER (WHERE exited = FALSE) AS retained_count, count(*) as total_customers,
 ROUND((COUNT(*) FILTER (WHERE exited = TRUE) * 100.0 / COUNT(*)), 2) AS churn_percentage
FROM bank_churn WHERE estimatedsalary <= 300000 GROUP BY FLOOR(estimatedsalary / 10000), geography
ORDER BY churn_percentage desc;
```

|    | geography<br>character varying (20) | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |    | geography<br>character varying (20) | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|-------------------------------------|-------------------------------|-----------------------------|------------------------|--------------------------|---------------------------|-----------------------------|----|-------------------------------------|-------------------------------|-----------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1  | Germany                             | 40000                         | 49999                       | 52                     | 79                       | 131                       | 39.69                       | 33 | France                              | 110000                        | 119999                      | 41                     | 195                      | 236                       | 17.37                       |
| 2  | Germany                             | 0                             | 9999                        | 50                     | 77                       | 127                       | 39.37                       | 34 | Spain                               | 130000                        | 139999                      | 23                     | 110                      | 133                       | 17.29                       |
| 3  | Germany                             | 60000                         | 69999                       | 46                     | 73                       | 119                       | 38.66                       | 35 | Spain                               | 110000                        | 119999                      | 24                     | 115                      | 139                       | 17.27                       |
| 4  | Germany                             | 10000                         | 19999                       | 40                     | 70                       | 110                       | 36.36                       | 36 | France                              | 130000                        | 139999                      | 41                     | 200                      | 241                       | 17.01                       |
| 5  | Germany                             | 80000                         | 89999                       | 41                     | 72                       | 113                       | 36.28                       | 37 | France                              | 150000                        | 159999                      | 39                     | 191                      | 230                       | 16.96                       |
| 6  | Germany                             | 110000                        | 119999                      | 41                     | 75                       | 116                       | 35.34                       | 38 | France                              | 180000                        | 189999                      | 39                     | 196                      | 235                       | 16.60                       |
| 7  | Germany                             | 180000                        | 189999                      | 45                     | 83                       | 128                       | 35.16                       | 39 | Spain                               | 100000                        | 109999                      | 21                     | 106                      | 127                       | 16.54                       |
| 8  | Germany                             | 140000                        | 149999                      | 43                     | 86                       | 129                       | 33.33                       | 40 | Spain                               | 30000                         | 39999                       | 18                     | 91                       | 109                       | 16.51                       |
| 9  | Germany                             | 160000                        | 169999                      | 41                     | 83                       | 124                       | 33.06                       | 41 | France                              | 100000                        | 109999                      | 44                     | 223                      | 267                       | 16.48                       |
| 10 | Germany                             | 20000                         | 29999                       | 42                     | 87                       | 129                       | 32.56                       | 42 | France                              | 10000                         | 19999                       | 37                     | 190                      | 227                       | 16.30                       |
| 11 | Germany                             | 120000                        | 129999                      | 40                     | 85                       | 125                       | 32.00                       | 43 | France                              | 160000                        | 169999                      | 40                     | 208                      | 248                       | 16.13                       |
| 12 | Germany                             | 30000                         | 39999                       | 36                     | 83                       | 119                       | 30.25                       | 44 | Spain                               | 0                             | 9999                        | 20                     | 104                      | 124                       | 16.13                       |
| 13 | Germany                             | 90000                         | 99999                       | 35                     | 81                       | 116                       | 30.17                       | 45 | France                              | 70000                         | 79999                       | 43                     | 224                      | 267                       | 16.10                       |
| 14 | Germany                             | 130000                        | 139999                      | 38                     | 90                       | 128                       | 29.69                       | 46 | Spain                               | 20000                         | 29999                       | 18                     | 95                       | 113                       | 15.93                       |
| 15 | Germany                             | 190000                        | 199999                      | 41                     | 100                      | 141                       | 29.06                       | 47 | France                              | 140000                        | 149999                      | 42                     | 224                      | 266                       | 15.79                       |
| 16 | Germany                             | 70000                         | 79999                       | 38                     | 93                       | 131                       | 29.01                       | 48 | France                              | 30000                         | 39999                       | 38                     | 211                      | 249                       | 15.26                       |
| 17 | Germany                             | 170000                        | 179999                      | 42                     | 106                      | 148                       | 28.38                       | 49 | Spain                               | 180000                        | 189999                      | 17                     | 96                       | 113                       | 15.04                       |
| 18 | Germany                             | 100000                        | 109999                      | 40                     | 102                      | 142                       | 28.17                       | 50 | Spain                               | 90000                         | 99999                       | 18                     | 110                      | 128                       | 14.73                       |
| 19 | Germany                             | 150000                        | 159999                      | 27                     | 69                       | 96                        | 28.13                       | 51 | Spain                               | 190000                        | 199999                      | 17                     | 99                       | 116                       | 14.66                       |
| 20 | Germany                             | 50000                         | 59999                       | 36                     | 101                      | 137                       | 26.28                       | 52 | France                              | 120000                        | 129999                      | 37                     | 221                      | 258                       | 14.34                       |
| 21 | Spain                               | 170000                        | 179999                      | 27                     | 94                       | 121                       | 22.31                       | 53 | Spain                               | 10000                         | 19999                       | 20                     | 121                      | 141                       | 14.18                       |
| 22 | Spain                               | 150000                        | 159999                      | 30                     | 105                      | 135                       | 22.22                       | 54 | Spain                               | 70000                         | 79999                       | 17                     | 105                      | 122                       | 13.93                       |
| 23 | France                              | 170000                        | 179999                      | 54                     | 208                      | 262                       | 20.61                       | 55 | Spain                               | 120000                        | 129999                      | 17                     | 105                      | 122                       | 13.93                       |
| 24 | Spain                               | 140000                        | 149999                      | 25                     | 101                      | 126                       | 19.84                       | 56 | Spain                               | 50000                         | 59999                       | 17                     | 107                      | 124                       | 13.71                       |
| 25 | Spain                               | 160000                        | 169999                      | 21                     | 85                       | 106                       | 19.81                       | 57 | Spain                               | 40000                         | 49999                       | 16                     | 112                      | 128                       | 12.50                       |
| 26 | Spain                               | 80000                         | 89999                       | 25                     | 111                      | 137                       | 18.99                       | 58 | France                              | 80000                         | 89999                       | 32                     | 228                      | 260                       | 12.31                       |
| 27 | France                              | 190000                        | 199999                      | 47                     | 205                      | 252                       | 18.65                       | 59 | France                              | 0                             | 9999                        | 31                     | 226                      | 257                       | 12.06                       |
| 28 | Spain                               | 60000                         | 69999                       | 20                     | 92                       | 112                       | 17.86                       | 60 | France                              | 40000                         | 49999                       | 27                     | 212                      | 239                       | 11.30                       |

|    | geography<br>character varying (20) | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint |    |         |        |        |    |     |
|----|-------------------------------------|-------------------------------|-----------------------------|------------------------|------------------------|----|---------|--------|--------|----|-----|
| 1  | France                              | 170000                        | 179999                      | 54                     | 208                    | 16 | Germany | 170000 | 179999 | 42 | 106 |
| 2  | Germany                             | 40000                         | 49999                       | 52                     | 79                     | 17 | France  | 110000 | 119999 | 41 | 195 |
| 3  | Germany                             | 0                             | 9999                        | 50                     | 77                     | 18 | Germany | 160000 | 169999 | 41 | 83  |
| 4  | France                              | 60000                         | 69999                       | 48                     | 228                    | 19 | Germany | 80000  | 89999  | 41 | 72  |
| 5  | France                              | 190000                        | 199999                      | 47                     | 205                    | 20 | Germany | 110000 | 119999 | 41 | 75  |
| 6  | Germany                             | 60000                         | 69999                       | 46                     | 73                     | 21 | Germany | 190000 | 199999 | 41 | 100 |
| 7  | Germany                             | 180000                        | 189999                      | 45                     | 83                     | 22 | France  | 130000 | 139999 | 41 | 200 |
| 8  | France                              | 100000                        | 109999                      | 44                     | 223                    | 23 | Germany | 100000 | 109999 | 40 | 102 |
| 9  | France                              | 20000                         | 29999                       | 44                     | 206                    | 24 | Germany | 10000  | 19999  | 40 | 70  |
| 10 | France                              | 90000                         | 99999                       | 43                     | 204                    | 25 | France  | 160000 | 169999 | 40 | 208 |
| 11 | Germany                             | 140000                        | 149999                      | 43                     | 86                     | 26 | Germany | 120000 | 129999 | 40 | 85  |
| 12 | France                              | 50000                         | 59999                       | 43                     | 204                    | 27 | France  | 150000 | 159999 | 39 | 191 |
| 13 | France                              | 70000                         | 79999                       | 43                     | 224                    | 28 | France  | 180000 | 189999 | 39 | 196 |
| 14 | Germany                             | 20000                         | 29999                       | 42                     | 87                     | 29 | Germany | 70000  | 79999  | 38 | 93  |
| 15 | France                              | 140000                        | 149999                      | 42                     | 224                    | 30 | Germany | 130000 | 139999 | 38 | 90  |

## Geography + Balance

SELECT geography, FLOOR(balance / 10000) \* 10000 AS balance\_range\_start, FLOOR(balance / 10000) \* 10000 + 9999 AS balance\_range\_end, COUNT(\*) FILTER (WHERE exited = TRUE) AS exited\_count, COUNT(\*) FILTER (WHERE exited = FALSE) AS retained\_count, COUNT(\*) AS total\_customers, ROUND((COUNT(\*) FILTER (WHERE exited = TRUE) \* 100.0 / COUNT(\*)), 2) AS churn\_percentage FROM bank\_churn WHERE balance <= 300000 GROUP BY FLOOR(balance / 10000), geography ORDER BY churn\_percentage desc;

|    | geography<br>character varying (20) | balance_range_start<br>numeric | balance_range_end<br>numeric | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |    | geography<br>character varying (20) | balance_range_start<br>numeric | balance_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint |
|----|-------------------------------------|--------------------------------|------------------------------|------------------------|--------------------------|---------------------------|-----------------------------|----|-------------------------------------|--------------------------------|------------------------------|------------------------|------------------------|
| 1  | France                              | 230000                         | 239999                       | 1                      | 0                        | 1                         | 100.00                      | 1  | France                              | 0                              | 9999                         | 338                    | 2081                   |
| 2  | France                              | 20000                          | 29999                        | 3                      | 0                        | 3                         | 100.00                      | 2  | Germany                             | 110000                         | 119999                       | 172                    | 199                    |
| 3  | Spain                               | 210000                         | 219999                       | 2                      | 0                        | 2                         | 100.00                      | 3  | Spain                               | 0                              | 9999                         | 163                    | 1036                   |
| 4  | Spain                               | 250000                         | 259999                       | 1                      | 0                        | 1                         | 100.00                      | 4  | Germany                             | 120000                         | 129999                       | 146                    | 237                    |
| 5  | Spain                               | 200000                         | 209999                       | 5                      | 2                        | 7                         | 71.43                       | 5  | Germany                             | 100000                         | 109999                       | 144                    | 208                    |
| 6  | France                              | 200000                         | 209999                       | 7                      | 3                        | 10                        | 70.00                       | 6  | Germany                             | 130000                         | 139999                       | 109                    | 205                    |
| 7  | Spain                               | 20000                          | 29999                        | 2                      | 1                        | 3                         | 66.67                       | 7  | Germany                             | 140000                         | 149999                       | 75                     | 157                    |
| 8  | Spain                               | 30000                          | 39999                        | 1                      | 1                        | 2                         | 50.00                       | 8  | Germany                             | 90000                          | 99999                        | 68                     | 173                    |
| 9  | Spain                               | 180000                         | 189999                       | 8                      | 8                        | 16                        | 50.00                       | 9  | France                              | 120000                         | 129999                       | 56                     | 285                    |
| 10 | Spain                               | 10000                          | 19999                        | 1                      | 1                        | 2                         | 50.00                       | 10 | France                              | 130000                         | 139999                       | 51                     | 233                    |
| 11 | Spain                               | 220000                         | 229999                       | 1                      | 1                        | 2                         | 50.00                       | 11 | France                              | 110000                         | 119999                       | 45                     | 273                    |
| 12 | Spain                               | 50000                          | 59999                        | 8                      | 9                        | 17                        | 47.06                       | 12 | France                              | 140000                         | 149999                       | 43                     | 203                    |
| 13 | Germany                             | 110000                         | 119999                       | 172                    | 199                      | 371                       | 46.36                       | 13 | France                              | 100000                         | 109999                       | 43                     | 244                    |
| 14 | France                              | 40000                          | 49999                        | 13                     | 16                       | 29                        | 44.83                       | 14 | Germany                             | 150000                         | 159999                       | 38                     | 107                    |
| 15 | France                              | 30000                          | 39999                        | 3                      | 4                        | 7                         | 42.86                       | 15 | France                              | 90000                          | 99999                        | 33                     | 204                    |
| 16 | Spain                               | 60000                          | 69999                        | 17                     | 23                       | 40                        | 42.50                       | 16 | France                              | 80000                          | 89999                        | 31                     | 151                    |
| 17 | Germany                             | 100000                         | 109999                       | 144                    | 208                      | 352                       | 40.91                       | 17 | France                              | 150000                         | 159999                       | 30                     | 130                    |
| 18 | France                              | 210000                         | 219999                       | 2                      | 3                        | 5                         | 40.00                       | 18 | Spain                               | 110000                         | 119999                       | 23                     | 120                    |
| 19 | Germany                             | 120000                         | 129999                       | 146                    | 237                      | 383                       | 38.12                       | 19 | Spain                               | 130000                         | 139999                       | 23                     | 113                    |
| 20 | France                              | 190000                         | 199999                       | 8                      | 15                       | 23                        | 34.78                       | 20 | Spain                               | 120000                         | 129999                       | 23                     | 151                    |
| 21 | Germany                             | 130000                         | 139999                       | 109                    | 205                      | 314                       | 34.71                       | 21 | Germany                             | 80000                          | 89999                        | 23                     | 97                     |
| 22 | France                              | 180000                         | 189999                       | 13                     | 26                       | 39                        | 33.33                       | 22 | France                              | 160000                         | 169999                       | 22                     | 102                    |
| 23 | Germany                             | 140000                         | 149999                       | 75                     | 157                      | 232                       | 32.33                       | 23 | France                              | 170000                         | 179999                       | 22                     | 48                     |
| 24 | France                              | 170000                         | 179999                       | 22                     | 48                       | 70                        | 31.43                       | 24 | Spain                               | 90000                          | 99999                        | 22                     | 99                     |
| 25 | Germany                             | 90000                          | 99999                        | 68                     | 173                      | 241                       | 28.22                       | 25 | Spain                               | 80000                          | 89999                        | 20                     | 78                     |
| 26 | Spain                               | 170000                         | 179999                       | 13                     | 35                       | 48                        | 27.08                       | 26 | France                              | 70000                          | 79999                        | 20                     | 99                     |
| 27 | France                              | 50000                          | 59999                        | 10                     | 27                       | 37                        | 27.03                       | 27 | Germany                             | 160000                         | 169999                       | 19                     | 64                     |
|    |                                     |                                |                              |                        |                          |                           |                             | 28 | Spain                               | 100000                         | 109999                       | 19                     | 128                    |
|    |                                     |                                |                              |                        |                          |                           |                             | 29 | Spain                               | 60000                          | 69999                        | 17                     | 23                     |
|    |                                     |                                |                              |                        |                          |                           |                             | 30 | Spain                               | 150000                         | 159999                       | 17                     | 64                     |

## Geography + Balance + Estimated Salary

Showing rows: 1 to 940

|   | geography<br>character varying (20) | balance_range_start<br>numeric | balance_range_end<br>numeric | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|---|-------------------------------------|--------------------------------|------------------------------|-------------------------------|-----------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 1 | Spain                               | 70000                          | 79999                        | 80000                         | 89999                       | 1                      | 0                        | 1                         | 100.00                      |
| 2 | France                              | 200000                         | 209999                       | 0                             | 9999                        | 1                      | 0                        | 1                         | 100.00                      |
| 3 | France                              | 30000                          | 39999                        | 30000                         | 39999                       | 1                      | 0                        | 1                         | 100.00                      |
| 4 | Germany                             | 160000                         | 169999                       | 150000                        | 159999                      | 1                      | 0                        | 1                         | 100.00                      |
| 5 | Spain                               | 20000                          | 29999                        | 60000                         | 69999                       | 1                      | 0                        | 1                         | 100.00                      |
| 6 | France                              | 50000                          | 59999                        | 110000                        | 119999                      | 1                      | 0                        | 1                         | 100.00                      |
| 7 | France                              | 200000                         | 209999                       | 130000                        | 139999                      | 1                      | 0                        | 1                         | 100.00                      |
| 8 | France                              | 40000                          | 49999                        | 180000                        | 189999                      | 1                      | 0                        | 1                         | 100.00                      |
| 9 | Spain                               | 250000                         | 259999                       | 80000                         | 89999                       | 1                      | 0                        | 1                         | 100.00                      |

|    | geography<br>character varying (20) | balance_range_start<br>numeric | balance_range_end<br>numeric | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | retained_count<br>bigint | total_customers<br>bigint | churn_percentage<br>numeric |
|----|-------------------------------------|--------------------------------|------------------------------|-------------------------------|-----------------------------|------------------------|--------------------------|---------------------------|-----------------------------|
| 56 | Spain                               | 60000                          | 69999                        | 70000                         | 79999                       | 3                      | 1                        | 4                         | 75.00                       |
| 57 | France                              | 70000                          | 79999                        | 100000                        | 109999                      | 3                      | 1                        | 4                         | 75.00                       |
| 58 | Spain                               | 70000                          | 79999                        | 20000                         | 29999                       | 3                      | 1                        | 4                         | 75.00                       |
| 59 | Germany                             | 110000                         | 119999                       | 40000                         | 49999                       | 14                     | 6                        | 20                        | 70.00                       |
| 60 | France                              | 50000                          | 59999                        | 80000                         | 89999                       | 2                      | 1                        | 3                         | 66.67                       |
| 61 | France                              | 60000                          | 69999                        | 160000                        | 169999                      | 2                      | 1                        | 3                         | 66.67                       |
| 62 | Spain                               | 180000                         | 189999                       | 180000                        | 189999                      | 2                      | 1                        | 3                         | 66.67                       |
| 63 | Germany                             | 70000                          | 79999                        | 110000                        | 119999                      | 2                      | 1                        | 3                         | 66.67                       |
| 64 | Spain                               | 60000                          | 69999                        | 60000                         | 69999                       | 2                      | 1                        | 3                         | 66.67                       |
| 65 | France                              | 40000                          | 49999                        | 120000                        | 129999                      | 2                      | 1                        | 3                         | 66.67                       |
| 66 | France                              | 190000                         | 199999                       | 130000                        | 139999                      | 2                      | 1                        | 3                         | 66.67                       |
| 67 | Germany                             | 160000                         | 169999                       | 80000                         | 89999                       | 2                      | 1                        | 3                         | 66.67                       |
| 68 | France                              | 30000                          | 39999                        | 170000                        | 179999                      | 2                      | 1                        | 3                         | 66.67                       |
| 69 | Germany                             | 100000                         | 109999                       | 60000                         | 69999                       | 10                     | 5                        | 15                        | 66.67                       |
| 70 | Spain                               | 90000                          | 99999                        | 160000                        | 169999                      | 2                      | 1                        | 3                         | 66.67                       |
| 71 | France                              | 190000                         | 199999                       | 50000                         | 59999                       | 2                      | 1                        | 3                         | 66.67                       |
| 72 | Germany                             | 140000                         | 149999                       | 60000                         | 69999                       | 8                      | 5                        | 13                        | 61.54                       |
| 73 | Germany                             | 130000                         | 139999                       | 0                             | 9999                        | 11                     | 7                        | 18                        | 61.11                       |
| 74 | Germany                             | 140000                         | 149999                       | 40000                         | 49999                       | 6                      | 4                        | 10                        | 60.00                       |
| 75 | France                              | 170000                         | 179999                       | 160000                        | 169999                      | 3                      | 2                        | 5                         | 60.00                       |
| 76 | Germany                             | 80000                          | 89999                        | 80000                         | 89999                       | 3                      | 2                        | 5                         | 60.00                       |
| 77 | Germany                             | 90000                          | 99999                        | 110000                        | 119999                      | 6                      | 4                        | 10                        | 60.00                       |
| 78 | Spain                               | 90000                          | 99999                        | 140000                        | 149999                      | 3                      | 2                        | 5                         | 60.00                       |
| 79 | Spain                               | 150000                         | 159999                       | 140000                        | 149999                      | 3                      | 2                        | 5                         | 60.00                       |
| 80 | France                              | 170000                         | 179999                       | 60000                         | 69999                       | 3                      | 2                        | 5                         | 60.00                       |
| 81 | Spain                               | 160000                         | 169999                       | 150000                        | 159999                      | 3                      | 2                        | 5                         | 60.00                       |
| 82 | Germany                             | 110000                         | 119999                       | 0                             | 9999                        | 10                     | 7                        | 17                        | 58.82                       |
| 83 | Germany                             | 110000                         | 119999                       | 110000                        | 119999                      | 12                     | 9                        | 21                        | 57.14                       |
| 84 | Germany                             | 100000                         | 109999                       | 10000                         | 19999                       | 8                      | 6                        | 14                        | 57.14                       |

|    | geography<br>character varying (20) | balance_range_start<br>numeric | balance_range_end<br>numeric | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint |
|----|-------------------------------------|--------------------------------|------------------------------|-------------------------------|-----------------------------|------------------------|------------------------|
| 1  | France                              | 0                              | 9999                         | 170000                        | 179999                      | 26                     | 95                     |
| 2  | France                              | 0                              | 9999                         | 190000                        | 199999                      | 23                     | 99                     |
| 3  | France                              | 0                              | 9999                         | 10000                         | 19999                       | 22                     | 95                     |
| 4  | France                              | 0                              | 9999                         | 70000                         | 79999                       | 21                     | 97                     |
| 5  | France                              | 0                              | 9999                         | 140000                        | 149999                      | 20                     | 112                    |
| 6  | France                              | 0                              | 9999                         | 100000                        | 109999                      | 20                     | 104                    |
| 7  | France                              | 0                              | 9999                         | 110000                        | 119999                      | 19                     | 92                     |
| 8  | France                              | 0                              | 9999                         | 90000                         | 99999                       | 19                     | 112                    |
| 9  | France                              | 0                              | 9999                         | 50000                         | 59999                       | 19                     | 105                    |
| 10 | France                              | 0                              | 9999                         | 130000                        | 139999                      | 18                     | 102                    |
| 11 | France                              | 0                              | 9999                         | 180000                        | 189999                      | 16                     | 94                     |
| 12 | France                              | 0                              | 9999                         | 20000                         | 29999                       | 15                     | 100                    |
| 13 | Germany                             | 110000                         | 119999                       | 40000                         | 49999                       | 14                     | 6                      |
| 14 | France                              | 0                              | 9999                         | 60000                         | 69999                       | 14                     | 106                    |
| 15 | France                              | 0                              | 9999                         | 80000                         | 89999                       | 14                     | 113                    |
| 16 | France                              | 0                              | 9999                         | 120000                        | 129999                      | 14                     | 105                    |
| 17 | Germany                             | 100000                         | 109999                       | 180000                        | 189999                      | 13                     | 11                     |
| 18 | Spain                               | 0                              | 9999                         | 80000                         | 89999                       | 13                     | 55                     |
| 19 | Spain                               | 0                              | 9999                         | 30000                         | 39999                       | 13                     | 42                     |
| 20 | France                              | 0                              | 9999                         | 40000                         | 49999                       | 13                     | 104                    |
| 21 | France                              | 0                              | 9999                         | 160000                        | 169999                      | 12                     | 107                    |
| 22 | France                              | 0                              | 9999                         | 0                             | 9999                        | 12                     | 132                    |
| 23 | Germany                             | 110000                         | 119999                       | 190000                        | 199999                      | 12                     | 13                     |
| 24 | Spain                               | 0                              | 9999                         | 170000                        | 179999                      | 12                     | 51                     |
| 25 | Germany                             | 110000                         | 119999                       | 110000                        | 119999                      | 12                     | 9                      |
| 26 | Germany                             | 130000                         | 139999                       | 0                             | 9999                        | 11                     | 7                      |
| 27 | France                              | 0                              | 9999                         | 30000                         | 39999                       | 11                     | 102                    |
| 28 | Germany                             | 120000                         | 129999                       | 0                             | 9999                        | 11                     | 12                     |
| 29 | Germany                             | 120000                         | 129999                       | 60000                         | 69999                       | 11                     | 9                      |
| 30 | Germany                             | 110000                         | 119999                       | 120000                        | 129999                      | 11                     | 13                     |

## Geography + Gender + Age + Balance

|    | gender<br>character varying (10) | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | balance_range_start<br>numeric | balance_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint |
|----|----------------------------------|-------------------------------------|-------------------------------------|-----------------------------------|--------------------------------|------------------------------|------------------------|------------------------|
| 1  | Female                           | France                              | 40                                  | 49                                | 0                              | 9999                         | 89                     | 215                    |
| 2  | Female                           | France                              | 50                                  | 59                                | 0                              | 9999                         | 49                     | 33                     |
| 3  | Male                             | France                              | 40                                  | 49                                | 0                              | 9999                         | 49                     | 253                    |
| 4  | Female                           | France                              | 30                                  | 39                                | 0                              | 9999                         | 42                     | 424                    |
| 5  | Male                             | France                              | 50                                  | 59                                | 0                              | 9999                         | 41                     | 48                     |
| 6  | Male                             | Germany                             | 40                                  | 49                                | 100000                         | 109999                       | 32                     | 25                     |
| 7  | Female                           | Germany                             | 40                                  | 49                                | 110000                         | 119999                       | 32                     | 20                     |
| 8  | Female                           | Spain                               | 40                                  | 49                                | 0                              | 9999                         | 32                     | 103                    |
| 9  | Female                           | Germany                             | 40                                  | 49                                | 120000                         | 129999                       | 32                     | 25                     |
| 10 | Male                             | Germany                             | 40                                  | 49                                | 110000                         | 119999                       | 32                     | 23                     |
| 11 | Male                             | France                              | 30                                  | 39                                | 0                              | 9999                         | 30                     | 579                    |
| 12 | Female                           | Germany                             | 40                                  | 49                                | 100000                         | 109999                       | 29                     | 23                     |
| 13 | Female                           | Germany                             | 40                                  | 49                                | 130000                         | 139999                       | 28                     | 21                     |
| 14 | Female                           | Spain                               | 30                                  | 39                                | 0                              | 9999                         | 27                     | 220                    |
| 15 | Female                           | Spain                               | 50                                  | 59                                | 0                              | 9999                         | 25                     | 24                     |
| 16 | Male                             | Germany                             | 40                                  | 49                                | 120000                         | 129999                       | 25                     | 31                     |
| 17 | Female                           | Germany                             | 30                                  | 39                                | 100000                         | 109999                       | 24                     | 52                     |
| 18 | Male                             | Spain                               | 40                                  | 49                                | 0                              | 9999                         | 24                     | 134                    |
| 19 | Female                           | Germany                             | 50                                  | 59                                | 110000                         | 119999                       | 23                     | 7                      |
| 20 | Male                             | Germany                             | 50                                  | 59                                | 120000                         | 129999                       | 22                     | 4                      |
| 21 | Male                             | Spain                               | 30                                  | 39                                | 0                              | 9999                         | 22                     | 277                    |
| 22 | Male                             | Germany                             | 30                                  | 39                                | 110000                         | 119999                       | 22                     | 57                     |
| 23 | Female                           | Germany                             | 30                                  | 39                                | 110000                         | 119999                       | 21                     | 33                     |
| 24 | Male                             | Germany                             | 50                                  | 59                                | 110000                         | 119999                       | 21                     | 5                      |
| 25 | Female                           | Germany                             | 30                                  | 39                                | 120000                         | 129999                       | 20                     | 55                     |
| 26 | Male                             | Germany                             | 40                                  | 49                                | 140000                         | 149999                       | 17                     | 14                     |
| 27 | Female                           | Germany                             | 30                                  | 39                                | 130000                         | 139999                       | 17                     | 40                     |
| 28 | Female                           | Germany                             | 40                                  | 49                                | 90000                          | 99999                        | 17                     | 23                     |
| 29 | Male                             | Germany                             | 40                                  | 49                                | 130000                         | 139999                       | 16                     | 26                     |
| 30 | Female                           | Germany                             | 50                                  | 59                                | 100000                         | 109999                       | 16                     | 3                      |

## Geography + Gender + Age + Balance + Estimated Salary

|    | gender<br>character varying (10) | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | balance_range_start<br>numeric | balance_range_end<br>numeric | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint |
|----|----------------------------------|-------------------------------------|-------------------------------------|-----------------------------------|--------------------------------|------------------------------|-------------------------------|-----------------------------|------------------------|------------------------|
| 1  | Female                           | France                              | 40                                  | 49                                | 0                              | 9999                         | 110000                        | 119999                      | 9                      | 8                      |
| 2  | Female                           | France                              | 40                                  | 49                                | 0                              | 9999                         | 140000                        | 149999                      | 7                      | 12                     |
| 3  | Female                           | France                              | 40                                  | 49                                | 0                              | 9999                         | 80000                         | 89999                       | 7                      | 12                     |
| 4  | Female                           | France                              | 40                                  | 49                                | 0                              | 9999                         | 60000                         | 69999                       | 6                      | 15                     |
| 5  | Female                           | France                              | 40                                  | 49                                | 0                              | 9999                         | 170000                        | 179999                      | 6                      | 13                     |
| 6  | Female                           | France                              | 40                                  | 49                                | 0                              | 9999                         | 100000                        | 109999                      | 6                      | 14                     |
| 7  | Female                           | Germany                             | 40                                  | 49                                | 100000                         | 109999                       | 180000                        | 189999                      | 6                      | 2                      |
| 8  | Female                           | France                              | 40                                  | 49                                | 0                              | 9999                         | 70000                         | 79999                       | 6                      | 8                      |
| 9  | Male                             | France                              | 30                                  | 39                                | 0                              | 9999                         | 50000                         | 59999                       | 6                      | 25                     |
| 10 | Male                             | France                              | 50                                  | 59                                | 0                              | 9999                         | 100000                        | 109999                      | 6                      | 0                      |
| 11 | Female                           | Spain                               | 50                                  | 59                                | 0                              | 9999                         | 80000                         | 89999                       | 5                      | 2                      |
| 12 | Female                           | France                              | 40                                  | 49                                | 0                              | 9999                         | 160000                        | 169999                      | 5                      | 13                     |
| 13 | Male                             | France                              | 40                                  | 49                                | 0                              | 9999                         | 190000                        | 199999                      | 5                      | 10                     |
| 14 | Female                           | France                              | 50                                  | 59                                | 0                              | 9999                         | 90000                         | 99999                       | 5                      | 2                      |
| 15 | Female                           | France                              | 50                                  | 59                                | 0                              | 9999                         | 170000                        | 179999                      | 5                      | 2                      |
| 16 | Male                             | France                              | 40                                  | 49                                | 0                              | 9999                         | 100000                        | 109999                      | 5                      | 13                     |
| 17 | Female                           | France                              | 50                                  | 59                                | 0                              | 9999                         | 60000                         | 69999                       | 4                      | 1                      |
| 18 | Female                           | France                              | 40                                  | 49                                | 0                              | 9999                         | 130000                        | 139999                      | 4                      | 14                     |
| 19 | Female                           | Spain                               | 50                                  | 59                                | 0                              | 9999                         | 30000                         | 39999                       | 4                      | 4                      |
| 20 | Female                           | France                              | 30                                  | 39                                | 0                              | 9999                         | 70000                         | 79999                       | 4                      | 22                     |
| 21 | Male                             | Germany                             | 30                                  | 39                                | 120000                         | 129999                       | 20000                         | 29999                       | 4                      | 4                      |
| 22 | Female                           | Germany                             | 30                                  | 39                                | 110000                         | 119999                       | 40000                         | 49999                       | 4                      | 0                      |
| 23 | Female                           | Germany                             | 30                                  | 39                                | 110000                         | 119999                       | 110000                        | 119999                      | 4                      | 1                      |
| 24 | Male                             | France                              | 50                                  | 59                                | 0                              | 9999                         | 170000                        | 179999                      | 4                      | 3                      |
| 25 | Male                             | France                              | 40                                  | 49                                | 0                              | 9999                         | 130000                        | 139999                      | 4                      | 11                     |
| 26 | Female                           | France                              | 30                                  | 39                                | 0                              | 9999                         | 10000                         | 19999                       | 4                      | 19                     |
| 27 | Female                           | Germany                             | 30                                  | 39                                | 120000                         | 129999                       | 50000                         | 59999                       | 4                      | 4                      |
| 28 | Male                             | Germany                             | 40                                  | 49                                | 110000                         | 119999                       | 20000                         | 29999                       | 4                      | 0                      |
| 29 | Male                             | Germany                             | 40                                  | 49                                | 120000                         | 129999                       | 20000                         | 29999                       | 4                      | 1                      |
| 30 | Male                             | Spain                               | 30                                  | 39                                | 0                              | 9999                         | 170000                        | 179999                      | 4                      | 6                      |

## Geography (Spain) + Age + Estimated Salary

| Showing rows: 1 to 129 |                                     |                                     |                                   |                               |                             |                        |                        |                                          |  |  |
|------------------------|-------------------------------------|-------------------------------------|-----------------------------------|-------------------------------|-----------------------------|------------------------|------------------------|------------------------------------------|--|--|
|                        | geography<br>character varying (20) | age_range_start<br>double precision | age_range_end<br>double precision | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100) |  |  |
| 1                      | Spain                               | 60                                  | 69                                | 140000                        | 149999                      | 1                      | 0                      | 100.00                                   |  |  |
| 2                      | Spain                               | 60                                  | 69                                | 120000                        | 129999                      | 3                      | 0                      | 100.00                                   |  |  |
| 3                      | Spain                               | 50                                  | 59                                | 140000                        | 149999                      | 11                     | 2                      | 84.62                                    |  |  |
| 4                      | Spain                               | 50                                  | 59                                | 180000                        | 189999                      | 7                      | 2                      | 77.78                                    |  |  |
| 5                      | Spain                               | 60                                  | 69                                | 170000                        | 179999                      | 5                      | 2                      | 71.43                                    |  |  |
| 6                      | Spain                               | 50                                  | 59                                | 90000                         | 99999                       | 5                      | 2                      | 71.43                                    |  |  |
| 7                      | Spain                               | 50                                  | 59                                | 160000                        | 169999                      | 4                      | 2                      | 66.67                                    |  |  |
| 8                      | Spain                               | 50                                  | 59                                | 80000                         | 89999                       | 9                      | 5                      | 64.29                                    |  |  |
| 9                      | Spain                               | 50                                  | 59                                | 150000                        | 159999                      | 9                      | 6                      | 60.00                                    |  |  |
| 10                     | Spain                               | 50                                  | 59                                | 70000                         | 79999                       | 4                      | 3                      | 57.14                                    |  |  |
| 11                     | Spain                               | 60                                  | 69                                | 110000                        | 119999                      | 4                      | 3                      | 57.14                                    |  |  |
| 12                     | Spain                               | 50                                  | 59                                | 100000                        | 109999                      | 4                      | 3                      | 57.14                                    |  |  |
| 13                     | Spain                               | 50                                  | 59                                | 190000                        | 199999                      | 5                      | 5                      | 50.00                                    |  |  |
| 14                     | Spain                               | 70                                  | 79                                | 190000                        | 199999                      | 1                      | 1                      | 50.00                                    |  |  |
| 15                     | Spain                               | 50                                  | 59                                | 60000                         | 69999                       | 5                      | 5                      | 50.00                                    |  |  |
| 16                     | Spain                               | 50                                  | 59                                | 30000                         | 39999                       | 7                      | 7                      | 50.00                                    |  |  |
| 17                     | Spain                               | 50                                  | 59                                | 110000                        | 119999                      | 5                      | 5                      | 50.00                                    |  |  |
| 18                     | Spain                               | 60                                  | 69                                | 20000                         | 29999                       | 2                      | 2                      | 50.00                                    |  |  |
| 19                     | Spain                               | 60                                  | 69                                | 70000                         | 79999                       | 3                      | 4                      | 42.86                                    |  |  |
| 20                     | Spain                               | 70                                  | 79                                | 90000                         | 99999                       | 2                      | 3                      | 40.00                                    |  |  |
| 21                     | Spain                               | 40                                  | 49                                | 150000                        | 159999                      | 15                     | 23                     | 39.47                                    |  |  |
| 22                     | Spain                               | 40                                  | 49                                | 0                             | 9999                        | 8                      | 15                     | 34.78                                    |  |  |
| 23                     | Spain                               | 10                                  | 19                                | 50000                         | 59999                       | 1                      | 2                      | 33.33                                    |  |  |
| 24                     | Spain                               | 60                                  | 69                                | 0                             | 9999                        | 1                      | 2                      | 33.33                                    |  |  |
| 25                     | Spain                               | 50                                  | 59                                | 120000                        | 129999                      | 3                      | 6                      | 33.33                                    |  |  |
| 26                     | Spain                               | 40                                  | 49                                | 160000                        | 169999                      | 10                     | 21                     | 32.26                                    |  |  |
| 27                     | Spain                               | 50                                  | 59                                | 130000                        | 139999                      | 3                      | 7                      | 30.00                                    |  |  |
| 28                     | Spain                               | 40                                  | 49                                | 140000                        | 149999                      | 9                      | 22                     | 29.03                                    |  |  |
| 29                     | Spain                               | 60                                  | 69                                | 100000                        | 109999                      | 2                      | 5                      | 28.57                                    |  |  |
| 30                     | Spain                               | 60                                  | 69                                | 30000                         | 39999                       | 2                      | 5                      | 28.57                                    |  |  |

## Geography (Spain) + Balance + Estimated Salary

| Showing rows: 1 to 299 |                                     |                                |                              |                               |                             |                        |                        |                                          |  |  |
|------------------------|-------------------------------------|--------------------------------|------------------------------|-------------------------------|-----------------------------|------------------------|------------------------|------------------------------------------|--|--|
|                        | geography<br>character varying (20) | balance_range_start<br>numeric | balance_range_end<br>numeric | salary_range_start<br>numeric | salary_range_end<br>numeric | exited_count<br>bigint | stayed_count<br>bigint | exited percentage ((exit/exit+stay)*100) |  |  |
| 21                     | Spain                               | 210000                         | 219999                       | 70000                         | 79999                       | 2                      | 0                      | 100.00                                   |  |  |
| 22                     | Spain                               | 160000                         | 169999                       | 160000                        | 169999                      | 1                      | 0                      | 100.00                                   |  |  |
| 23                     | Spain                               | 200000                         | 209999                       | 150000                        | 159999                      | 2                      | 0                      | 100.00                                   |  |  |
| 24                     | Spain                               | 200000                         | 209999                       | 40000                         | 49999                       | 1                      | 0                      | 100.00                                   |  |  |
| 25                     | Spain                               | 20000                          | 29999                        | 60000                         | 69999                       | 1                      | 0                      | 100.00                                   |  |  |
| 26                     | Spain                               | 50000                          | 59999                        | 10000                         | 19999                       | 2                      | 0                      | 100.00                                   |  |  |
| 27                     | Spain                               | 60000                          | 69999                        | 70000                         | 79999                       | 3                      | 1                      | 75.00                                    |  |  |
| 28                     | Spain                               | 70000                          | 79999                        | 20000                         | 29999                       | 3                      | 1                      | 75.00                                    |  |  |
| 29                     | Spain                               | 150000                         | 159999                       | 100000                        | 109999                      | 3                      | 1                      | 75.00                                    |  |  |
| 30                     | Spain                               | 90000                          | 99999                        | 160000                        | 169999                      | 2                      | 1                      | 66.67                                    |  |  |
| 31                     | Spain                               | 180000                         | 189999                       | 180000                        | 189999                      | 2                      | 1                      | 66.67                                    |  |  |
| 32                     | Spain                               | 60000                          | 69999                        | 60000                         | 69999                       | 2                      | 1                      | 66.67                                    |  |  |
| 33                     | Spain                               | 160000                         | 169999                       | 150000                        | 159999                      | 3                      | 2                      | 60.00                                    |  |  |
| 34                     | Spain                               | 90000                          | 99999                        | 140000                        | 149999                      | 3                      | 2                      | 60.00                                    |  |  |
| 35                     | Spain                               | 150000                         | 159999                       | 140000                        | 149999                      | 3                      | 2                      | 60.00                                    |  |  |
| 36                     | Spain                               | 50000                          | 59999                        | 190000                        | 199999                      | 1                      | 1                      | 50.00                                    |  |  |
| 37                     | Spain                               | 110000                         | 119999                       | 180000                        | 189999                      | 3                      | 3                      | 50.00                                    |  |  |
| 38                     | Spain                               | 70000                          | 79999                        | 130000                        | 139999                      | 1                      | 1                      | 50.00                                    |  |  |
| 39                     | Spain                               | 120000                         | 129999                       | 170000                        | 179999                      | 4                      | 4                      | 50.00                                    |  |  |